GE Medical Systems

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Technical Publication

Direction 2299680-100
Revision 3

GE Medical Systems
Remote Tilt Upgrade

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

このサービスマニュアルには英語版しかありません。
ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
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注意:

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write “Damage In Shipment” on ALL copies of the freight or express bill BEFORE delivery is accepted or “signed for” by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage that may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	6/27/01	Initial release of document.
1	10/08/01	Sections 4, 5 and 6 updated for serviceability of remote tilt upgrade.
2	02/08/02	CQA 1020600: Addition of Sanota board and schematics.
3	05/14/02	General Updates to Procedure.

List of Effected Pages

PAGES	REVISION	PAGES	REVISION
1 through 72	1		
55 through 70	2		
21, 32, 35–37, 38–40, 41, 42, 43, 48	3		

Table of Contents

Preface
Publication Conventions

Purpose: Standardize conventions for representing information is a uniform way of communicating information to a reader in a consistent manner. Conventions are used so that the reader can easily recognize the actions or decisions that must be made. There are a number of character and paragraph styles used in this publication to accomplish this task. Please become familiar with them before proceeding forward.

Section 1.0
Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicated potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

It's important that you read and understand hazard statements, and not just ignore them.

DANGER SEVERE PERSONAL INJURY

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**

WARNING SERIOUS PERSONAL INJURY

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **POTENTIAL FOR SHOCK**
- **EXPOSED WIRES**
- **FAILURE TO TAG AND LOCKOUT SYSTEM POWER COULD ALLOW FOR UN-COMMAND MOTION.**

CAUTION Minor Personal Injury

Caution is used when a hazard exists which <u>can or could cause minor injury</u> to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**

NOTICE Property Damage Only

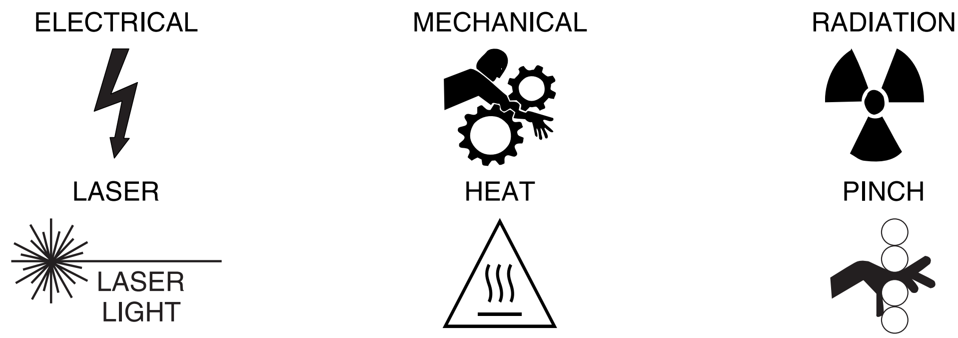
Notice is used when a hazard is present that can cause property damage but has <u>absolutely no personal injury risk</u>.They can include:

- **Disk drive will be destroyed**
- **Internal mechanical damage, such as to the x-ray tube**
- **Coasting the rotor through resonance.**

1.2 Graphical Representation

Important information will always be preceded by the exclamation point contained within a triangle, as seen throughout the Preface chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

HAZARDS



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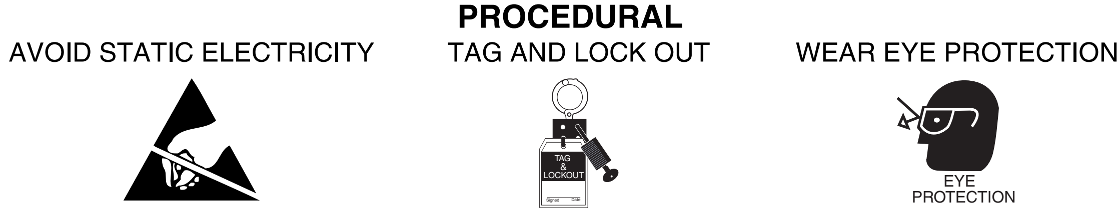
PROCEDURAL



Section 2.0
Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as purpose, example, comment or note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

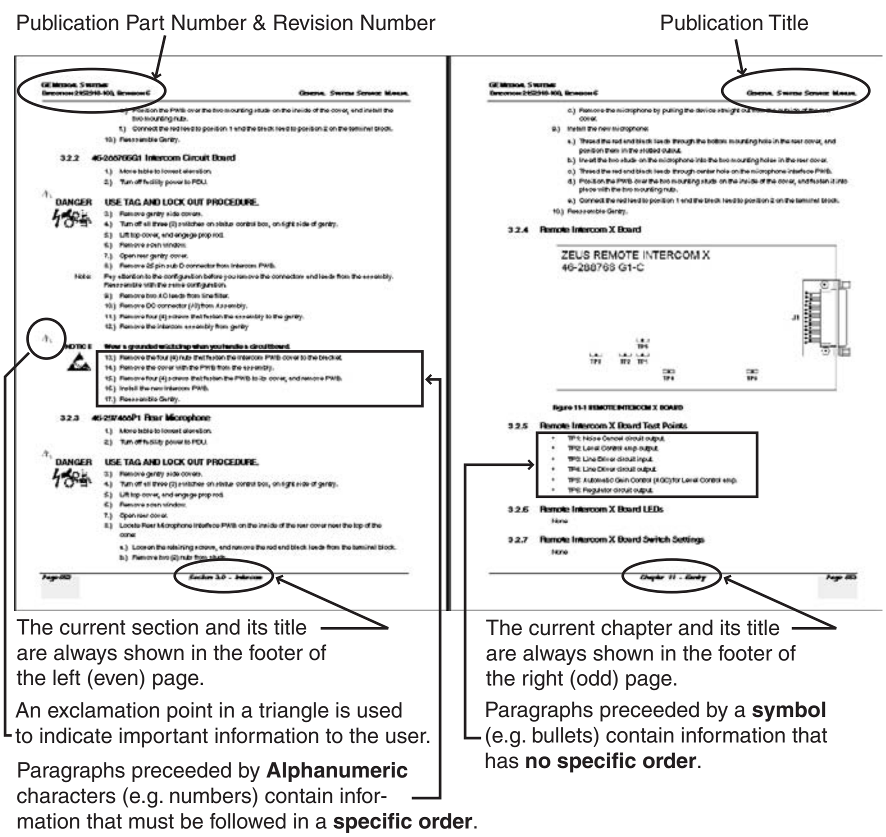
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant to your situation.

2.2 Page Layout



Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Text Character Styles

Within this publication, mono-spaced character styles (fonts) are used to indicate computer text that's either screen input and output. Mono spaced fonts, such courier, are used to indicated text direction. When you type at your keyboard, your are generating computer input. Occasionally you will see the math operators "greater than" and "less than" symbols used to indicate the start and finish of variable output. When reading text generated by the computer, you are reading it as computer generated output. In addition to direction, characters are italicized (e.g. *italics*) to indicate information specific to your system or site.

Example: Fixed Output — This paragraph's fonts represents computer generated screen "fixed" output. It's output is fixed from the sense that it does not vary from application to application. It's the most commonly used style used to indicate filenames, paths and text that does not change form system to system. The character style used is a fixed width such as courier.

Example: Variable Output — *This paragraph's font represents computer screen output that is "variable". Its used to represent output that varies from application to application or system to system. Variable output is sometimes found placed between greater than and lesser than operators for clarification. For example: <variable_ouput> or <3.45.120.3>. In both cases, the < and > operators are not part of the actual input.*

Example: Fixed Input — **This paragraph's font represents fixed input. It's computer input that is typed-in via the keyboard. Typed input that does not vary from application to application or system to system. Fixed text the user is required to supply as input. For example: cd /usr/3p**

Example: Variable Input — ***This paragraph's font represents computer input that can vary from application to application or system to system. With variable text, the user is required to supply system dependent input or information. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators would be dropped prior to input. For example:*** **ypcat hosts | grep** ***<3.45.120.3>*** **would be type into the computer as ypcat hosts | grep** ***3.45.120.3 without the greater than and less than operators.***

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys — A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Soft Keys — Whereas the computer MENU button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

Remote Tilt Upgrade

Purpose: This document defines the process to incorporate the Remote Tilt Option for either CT/i or LightSpeed systems.

Section 1.0 Overview

The Remote Tilt Upgrade is used on either the CT/i or LightSpeed systems. This section defines which systems this upgrade is applicable and time required to install.

1.1 Effectivity

The Remote Tilt Upgrade is for systems that contain an Octane computer within the console only. This upgrade is not compatible with systems that contain an Indigo computer within the console.

This upgrade is applicable to the following systems:

- CT/i Octane Software Revision 6.3 or Higher
- LightSpeed (QX/i) Software Revision 1.4 or Higher

1.2 Before You Begin

- Check all parts in the kit with the list in Section 7.0.
- Refer to the *Software Load Procedures* ("Load from Cold") and test the CD-ROM drive to ensure that it can read from the CD.

WARNING **MAKE SURE ALL LOCKOUT/TAGOUT PROCEDURES HAVE BEEN FOLLOWED BEFORE PERFORMING AN UPGRADE TO THESE SYSTEMS.**

1.2.1 Time Needed and Tools Required

PROCESS		TIME	SPECIAL TOOLS
Pre-installation Check		15 min.	
GANTRY	Remove Top and Side Covers	10 min.	
	Install Front Switch	15 min.	Drill w/Guide 2295236
	Install Front Cable	15 min.	
	Install Rear Switch	25 min.	Drill w/Guide 2295236
	Install Rear Cable	15 min.	
	Install Top and Side Covers	10 min.	

Table 1-1Time and Tools Required

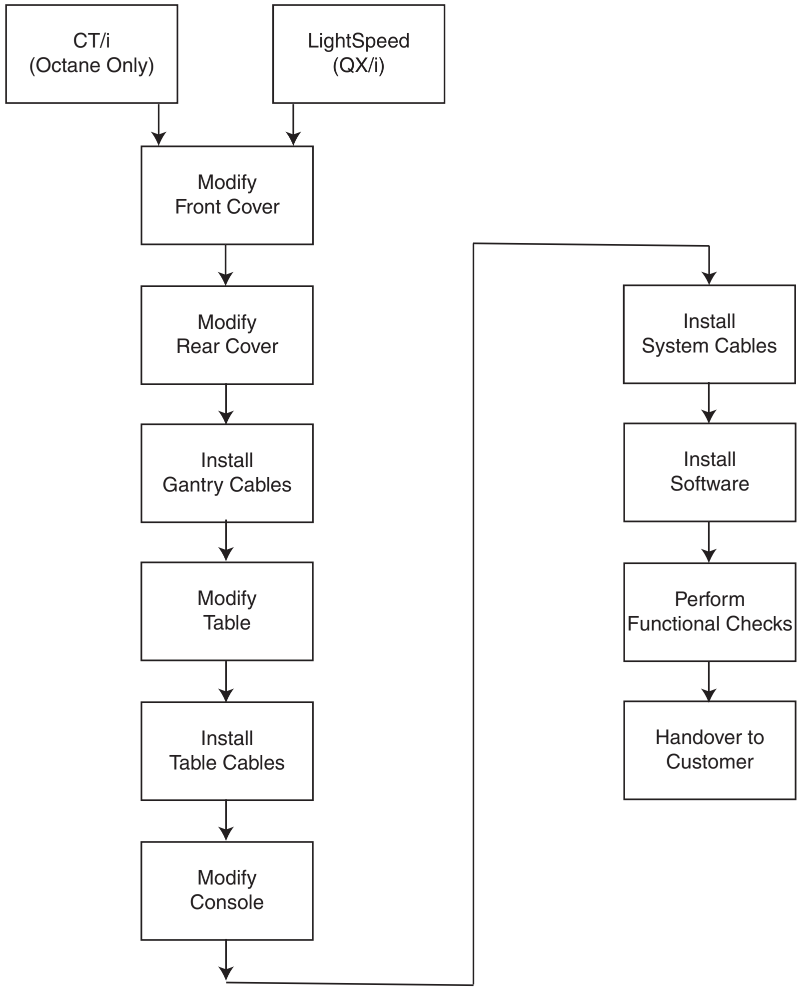
PROCESS		TIME	SPECIAL TOOLS
TABLE	Remove Side Covers	10 min.	
	Remove Existing ETC Board	15 min.	
	Install New ETC Board	15 min.	
	Install Cables	10 min.	
	Install Side Covers	10 min.	
CONSOLE	Remove Existing Keyboard	5 min.	
	Install SCIM and New Keyboard	10 min.	
	Remove Front Covers	5 min.	
	Remove Existing Intercom Board	15 min.	
	Install New Intercom Board	15 min.	
	Install Cables	2 hours	
	Install Upgrade Software	35 min.	
	Install Front Cover	5 min.	
Functional Checks		45 min.	
Clean-up		30 min.	
	Total Time	7.5 hours	

Table 1-1Time and Tools Required

Additional tools required to complete this upgrade:

- electric drill
- 10mm drill
- VOM meter
- vacuum cleaner

1.2.2 Work Flow Diagram



1.2.3 Referenced Documentation

The following referenced documents should be available during installation of the upgrade:

- *Remote Tilt Upgrade Installation Manual*, 2299680-100.
- *System Installation Manual*, 2152926-100.
- C*T/i Operator's Reference Manual*, 2142878-100.
- *Prescribed Tilt Operator Manual*, 2301300-100.

1.2.4 CTQs

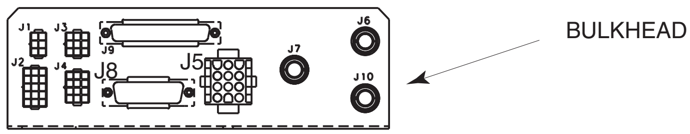
Critical to Quality (CTQ) are issues that have been identified as significant for satisfying the needs and expectations of our customers. These factors have usually been identified via a customer/ vendor survey.

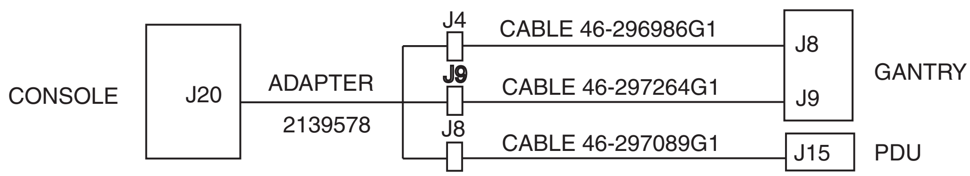
DANGER **THE POSITIONS OF THE FRONT AND REAR CONTACT SENSORS ARE CRITICAL TO ENSURE THAT THEY FUNCTION PROPERLY. THEY MUST BE PLACED IN THE TOP CENTER OF THE CONE ON BOTH FRONT AND REAR GANTRY COVERS.**

1.2.5 Configuration Identification

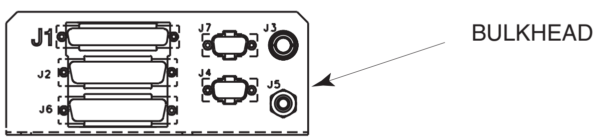
This Remote Tilt Upgrade is applicable to three system configurations. Refer to the gantry bulkhead, and its connecting cables, to determine system configuration.

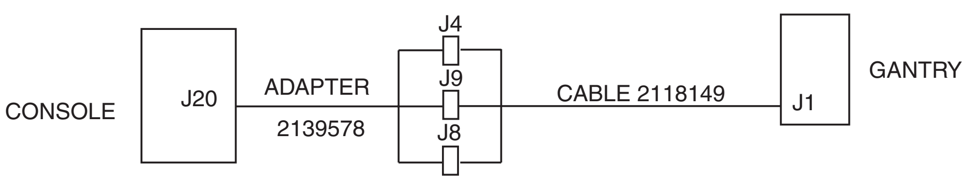
1.2.5.1 Configuration #1



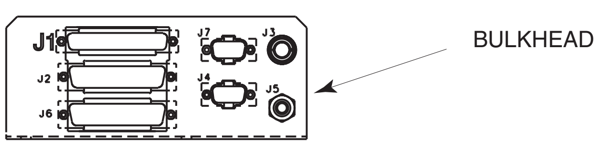


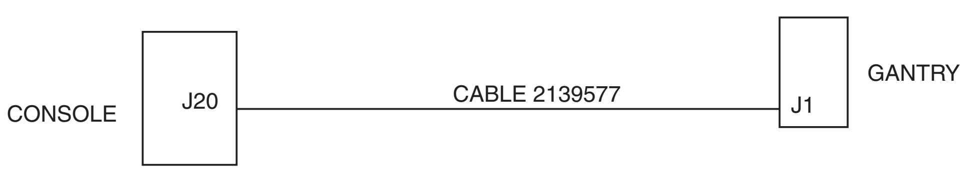
1.2.5.2 Configuration #2





1.2.5.3 Configuration #3





Section 2.0 Installation

Refer to system service documents for specific referenced procedures.

2.1 Remove Covers

WARNING MAKE SURE ALL LOCKOUT/TAGOUT PROCEDURES HAVE BEEN FOLLOWED BEFORE PERFORMING AN UPGRADE TO THESE SYSTEMS.

To gain access to internal components of the gantry, table base, and console, covers will need to be removed: Once the upgrade is complete, covers are installed in the reverse order.

2.1.1 Gantry

1.) Remove side covers from the gantry. Refer to *System Installation Manual*, 2152926-100.
2.) Remove top covers from the gantry. Refer to *System Installation Manual*, 2152926-100.

2.1.2 Table

1.) Remove right base cover from the table base. Refer to *System Installation Manual*, 2152926-100.
2.) Remove all covers between the table base and gantry base. Refer to *System Installation Manual*, 2152926-100.

2.1.3 Console

Removal front cover from the console. Refer to *System Installation Manual*, 2152926-100.

2.2 Gantry Modifications

A contact sensor and harness are installed on both front and rear covers. Each harness is connected to a new adapter cable from the gantry bulkhead and table base.

2.2.1 Front Cover

WARNING MAKE SURE ALL LOCKOUT/TAGOUT PROCEDURES HAVE BEEN FOLLOWED BEFORE PERFORMING AN UPGRADE TO THESE SYSTEMS.

2.2.1.1 Contact Sensor

The contact sensor detects when an object (such as a patient) comes within a specified range of the gantry cover.

1.) Remove palm screw from front cover.
2.) Swing open front cover from the gantry.
3.) Position contact sensor 2286908 on cone center of the cover (use microphone as a center guide), move the sensor up and down until curve of sensor matches curve of cover, and make sure sensor does not come in contact with microphone.

Note: Sensor may malfunction when microphone interferes with proper placement of tilt sensor assembly causing the tilt function to be inoperable. If microphone is interfering with the proper placement of the sensor, remove microphone from the cover, and mount microphone in new location according to 2.2.1.2.

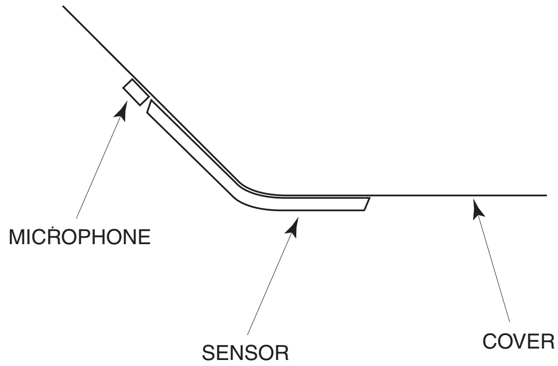


Figure 2-1 Mounting Sensor on Front Cover

4.) Tape sensor into position.
5.) Verify proper fit of sensor on the cover.
6.) Mark location of the sensor lead.
  - Put one piece of tape at the top edge of the connector.
  - Put one piece of tape under lead at point of sensor edge.
  - Mark the centerline on the tape at the end of the connector.
  - Mark locator lines on the tape at each side that the sensor lead and sensor edge intercept/meet.

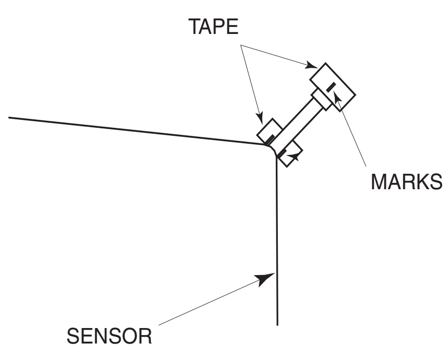


Figure 2-2 Marks For Lead Hole In Front Cover

7.) Place plastic large enough to cover cone bottom of front cover to prevent debris from falling into the gantry.
8.) Using switch mounting drill with guide 2295236 provided with the upgrade, drill pilot holes into the cover through each of the two mounting holes of the sensor.
9.) Remove sensor from the cover.
10.) Mark location of hole for sensor lead.
  - Align a straight edge so it is through the sensor connector centerline and between the two interception marks on the tape.
  - Draw a line along the straight edge from the sensor edge past the sensor pilot hole.
  - Put a mark 20mm from the edge of tape to the sensor lead centerline.

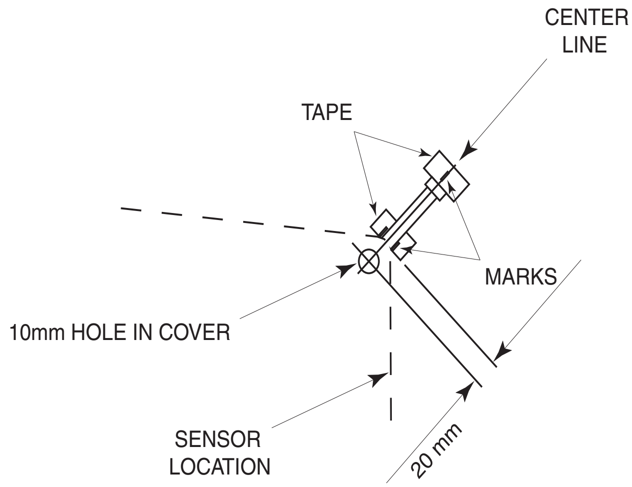


Figure 2-3 Position of Hole for Front Sensor Lead

11.) Using switch mounting drill with guide 2295236 provided with the upgrade, drill a hole into the cover through the location of hole for the sensor lead.
12.) Drill a 4.2mm hole through two pilot holes in the cover for each sensor.
13.) Drill a 10mm hole through pilot hole in the cover for the sensor lead.
14.) Vacuum all loose debris from inside and outside the front cover.
15.) Remove tape from the front cover.
16.) Clean sensor mounting area of the front cover.
17.) Remove plastic from cone of the front cover.

NOTICE **Failure to remove all debris from sensor before installation could result in sensor not functioning properly.**

18.) Vacuum all debris from the sensor.
19.) Put a bead of RTV 46-170619P1 around the inside edge of the sensor cover.

Note: Make sure bead of RTV is large enough to seal the sensor.

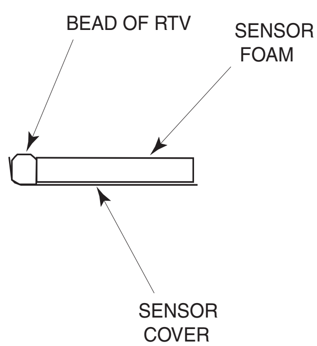


Figure 2-4 RTV Bead on Sensor Cover

20.) Put sensor lead through sensor lead hole, align two mounting holes of the sensor, and press sensor onto cover.
21.) Install two mounting screws 2286911 into sensor, taping each screw in place, and secure sensor to the cover using lockwashers and nuts.
22.) Tape sensor edges to the cover. (Tape can be removed once RTV has cured.)
23.) Using an Ohm meter (between pins 1 and 2 of connector), check sensor for proper operation.

- Normal circuit should read 20K +200/-0 Ohms
- Press on switch should result in a reading of less than 100 Ohms

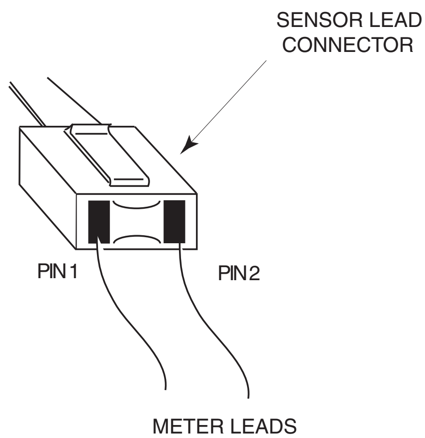


Figure 2-5 Sensor Pin Locations

2.2.1.2 Mount Microphone

If the microphone was not removed during sensor installation, this Mount Microphone procedure will not be required. However, if the microphone was removed to allow room for sensor to be mounted during the sensor installation procedure, perform the following.

1.) Place plastic large enough to cover cone bottom of gantry cover to prevent debris from falling into gantry.

Note: If the sensor does not cover any of the top microphone mounting holes on the cover, use the hole closest to the sensor as the new bottom mounting hole for the microphone.

2.) Locate the center of the cover above the sensor, use the microphone circuit board as a template, and drill three new vertical mounting holes for the microphone. (Use 3.3 mm drill for top and bottom holes for mounting microphone and 3.3 m drill for middle hole for microphone lead.

3.) Install microphone into cover by placing microphone lead through cover and securing microphone to cover with two nuts. (These are the same nuts removed during microphone removal.)

4.) Connect microphone lead to the harness.

2.2.1.3 Interference Harness

The interface harness provides the connection between the contact sensor and the adapter cable in the gantry.

1.) Connect harness 2295234 to sensor connector.

2.) Tape harness to cover. (Tape can be removed once RTV has cured.)

3.) Allowing for a service loop at the sensor connector, tie wrap front cover harness (approximately every 6 to 8 inches) up to the microphone cable, and then over to the cables running along the top of the cover to the STC.

4.) Apply RTV 46-170619P1 between cable and cover where it is not connected to a cable. (This will keep harness close to the cover so it does not catch when frame is rotating.)

2.2.2 Rear Cover

WARNING MAKE SURE ALL LOCKOUT/TAGOUT PROCEDURES HAVE BEEN FOLLOWED BEFORE PERFORMING AN UPGRADE TO THESE SYSTEMS.

2.2.2.1 Contact Sensor

The contact sensor detects when an object (such as a patient) comes within a specified range of the gantry cover.

1.) Remove four palm screws from the rear cover.
2.) Slide the rear cover back from the front cover.
3.) Position sensor 2286909 on bottom of cone center of the rear cover, and move the sensor up and down until curve of sensor matches curve of cover.
4.) Measure the distance from the sensor to the edge of the cone.
5.) Move the sensor to the top of the cone (use microphone as a center guide), placing the sensor the same distance from the cone edge measured on the bottom, and make sure sensor does not come in contact with microphone.

Note: Sensor may malfunction when microphone interferes with proper placement of tilt sensor assembly causing the tilt function to be inoperable. If microphone is interfering with the proper placement of the sensor, remove microphone from the cover, and mount microphone in new location according to 2.2.1.2.

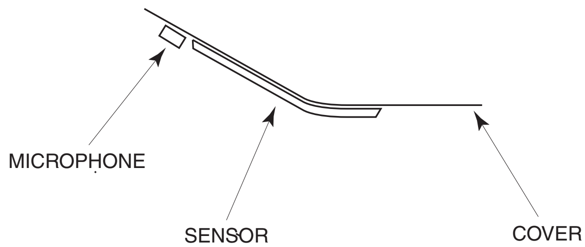


Figure 2-6 Mounting Sensor on Rear Cover

6.) Tape sensor into position.
7.) Verify proper fit of sensor on the cover.
8.) Mark location of the sensor lead.
   - Put one piece of tape on cover at the top edge of the connector.
   - Put one piece of tape on cover under lead at point of sensor edge.
   - Mark the centerline on the tape at the end of the connector.
   - Mark locator lines on the tape at each side that the sensor lead and sensor edge intercept/meet.

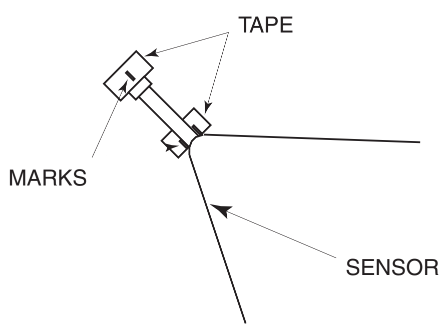


Figure 2-7 Marks For Lead Hole In Rear Cover

9.) Place plastic large enough to cover the cone bottom of rear cover to prevent debris from falling into the gantry.
10.) Using switch mounting drill with guide 2295236 provided with the upgrade, drill pilot holes into the cover through each of the six mounting holes of the sensor.
11.) Remove sensor from the cover.

12.) Mark location of hole for sensor lead.
- Align a straight edge so it is through the sensor connector centerline and between the two interception marks on the tape.
- Draw a line along the straight edge from the sensor edge past the sensor pilot hole.
- Put a mark 20mm from the edge of tape to the sensor lead centerline.

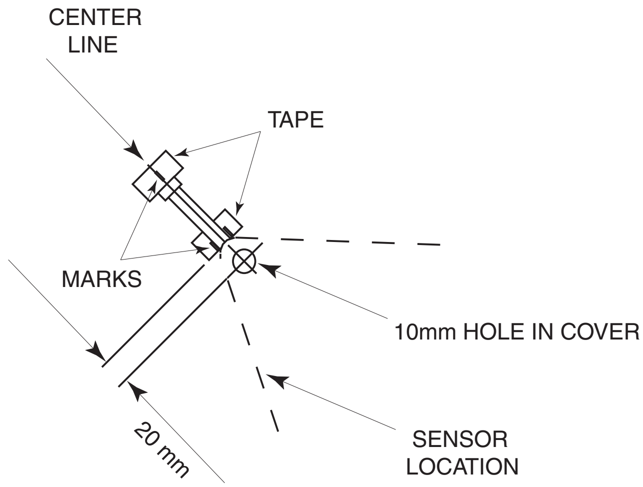


Figure 2-8 Position of Sensor Lead Hole

13.) Using switch mounting drill with guide 2295236 provided with the upgrade, drill a hole into the cover through the location of the sensor lead hole.
14.) Drill 4.2mm hole through six pilot holes in the cover for the sensor.
15.) Drill a 10mm hole through pilot hole in the cover for the sensor lead.
16.) Vacuum all loose debris from inside and outside of the rear cover.
17.) Remove tape from rear cover.
18.) Clean sensor mounting area of the cover.
19.) Vacuum all debris from the sensor.

Note: Failure to remove all debris from sensor before installation could result in sensor not functioning properly.

20.) Put a bead of RTV 46-170619P1 around the inside edge of the sensor cover.

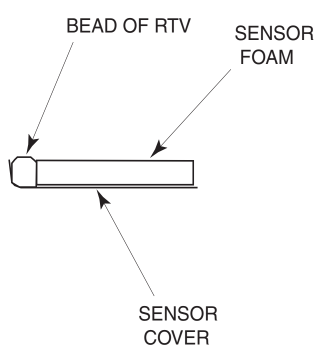


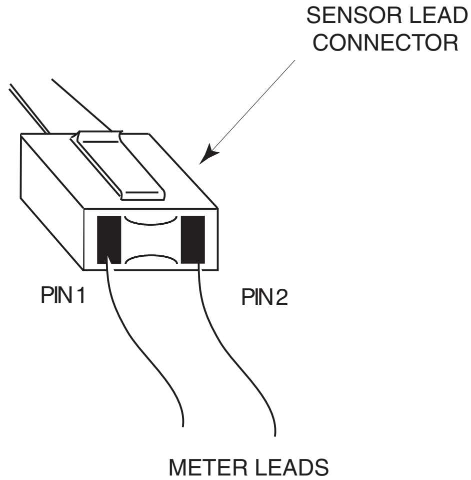
Figure 2-9 RTV Bead on Sensor Cover

21.) Put sensor lead through sensor lead hole, align six mounting holes of the sensor, and press sensor onto rear cover.
22.) Install six mounting screws 2286911 into sensor, taping each screw in place, and secure sensor to rear cover using nuts with lockwashers.

23.) Tape sensor edges to the cover. (Tape can be removed once RTV has cured.)

24.) Using an Ohm meter (between pin 1 and 2 of connector), check sensor for proper operation.

- Normal circuit should read 20K +200/-0 Ohms
- Press on switch should result in a reading of less than 100 Ohms



2.2.2.2 Interference Harness

The interface harness provides the connection between the tilt sensor assembly and the adapter cable in the gantry.

1.) Connect harness 2295235 to sensor connector.

2.) Tape harness to cover. (Tape can be removed once RTV has cured.)

3.) Allowing for a service loop at the sensor connector, tie wrap rear cover harness (approximately every 6 to 8 inches) up to the microphone cable, and then over to the cables running along the top of the cover to above the STC.

4.) Apply RTV 46-170619P1 between cable and cover where it is not connected to a cable. (This will keep harness close to the cover so it does not catch when frame is rotating.)

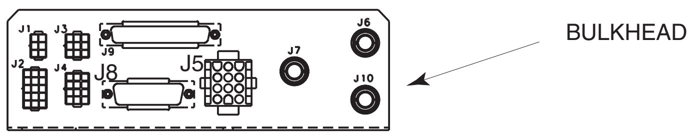
2.2.3 Adapter Cable

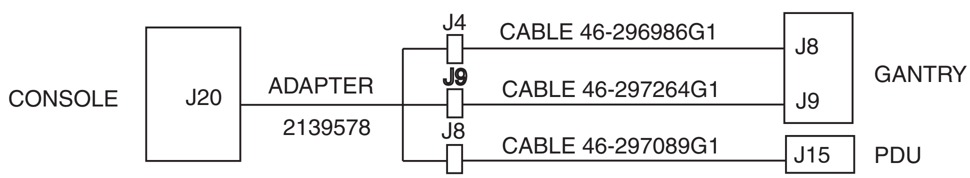
The adapter cable provides the connection between the interface harness and the bulkhead in the gantry.

2.2.3.1 Identify Configuration

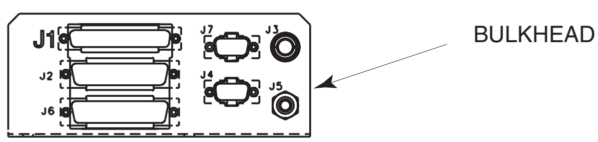
This Remote Tilt Upgrade is applicable to three system configurations. Refer to the gantry bulkhead, and its connecting cables, to determine system configuration. Compare the bulkhead and the J20 cable part numbers to identify the configuration.

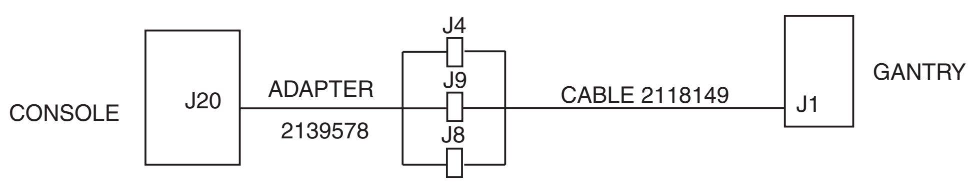
2.2.3.2 Configuration #1



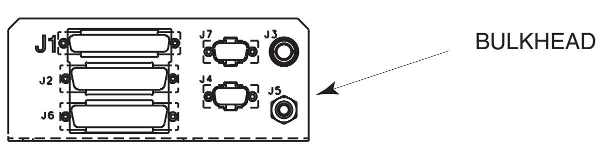


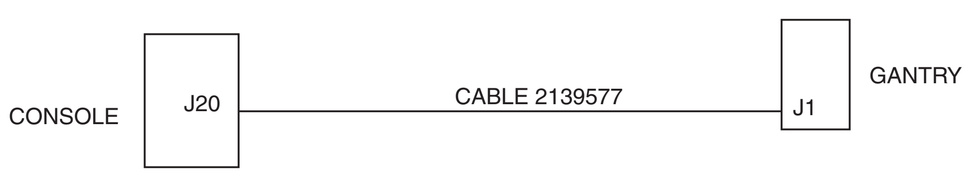
2.2.3.3 Configuration #2





2.2.3.4 Configuration #3





2.2.3.5 Installation

The adapter cable is installed in the gantry. It completes the connection between front and right sensors to the bulkhead, console, and table (see Figure 2-9). Refer to the specific configuration installation procedure for details.

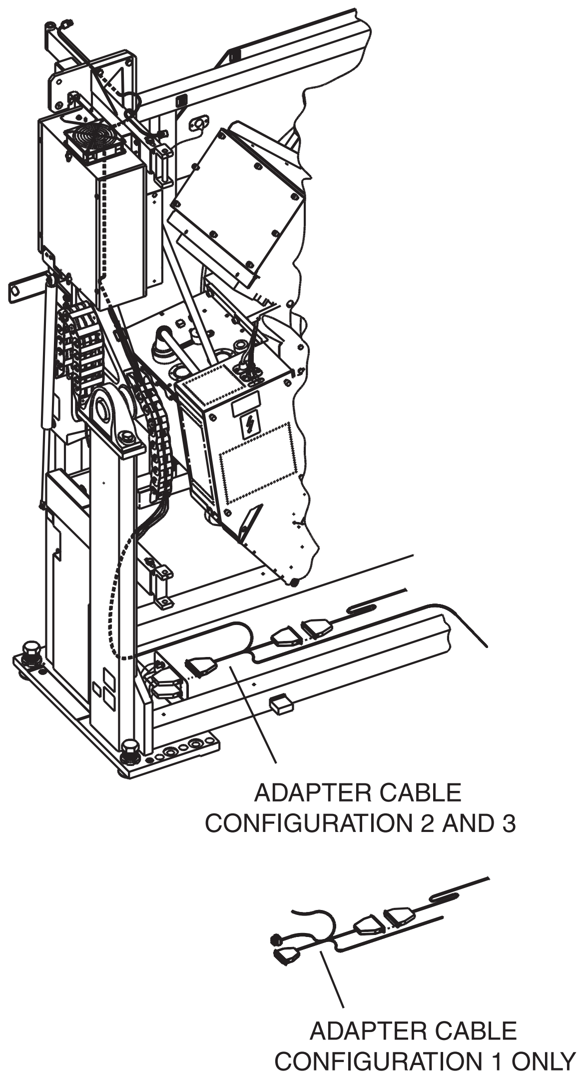
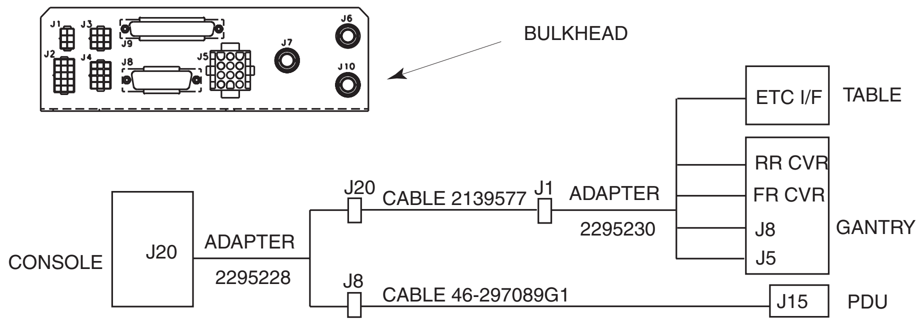


Figure 2-10 Adapter Cable in Gantry

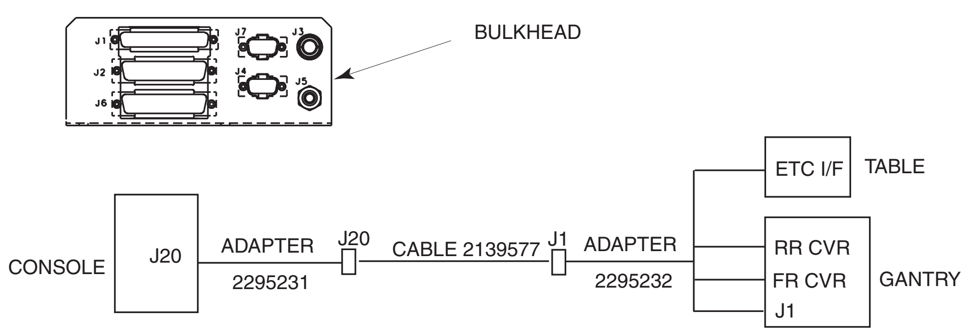
Configuration #1

1.) Disconnect cable 46-296986G1 from J8 on the gantry bulkhead.
2.) Disconnect cable 46-297264G1 from J5 on the gantry bulkhead.
3.) Feed adapter cable 2295230 (end with two connectors marked front and rear cover) up through the gantry.
    - Follow up existing harness along vertical channel.
    - Route along the outside of the track.
    - Front connector to front cover.
    - Rear connector to rear cover.
4.) Connect connectors J5 and J8 to the gantry bulkhead.
5.) Connect connectors to the front interference harness.
6.) Connect connectors to the rear interference harness.
7.) Feed long adapter cable end through horizontal cable channel to the table base.



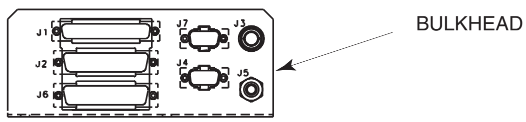
Configuration #2

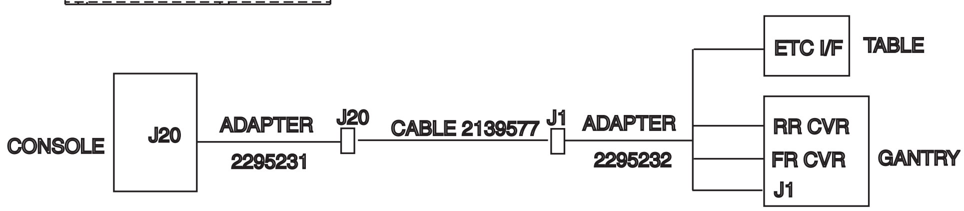
1.) Disconnect cable 2118149 from J1 on the gantry bulkhead.
2.) Feed adapter cable 2295232 (end with two connectors marked front and rear cover) up through the gantry.
    - Follow up existing harness along vertical channel.
    - Route along the outside of the track.
    - Front connector to front cover.
    - Rear connector to rear cover.
3.) Connect connector J1 to the gantry bulkhead.
4.) Connect connectors to the front interference harness.
5.) Connect connectors to the rear interference harness.
6.) Feed long adapter cable end through horizontal cable channel to the table base.



Configuration #3

1.) Disconnect cable 2139577 from J1 on the gantry bulkhead.
2.) Feed adapter cable 2295232 (end with two connectors marked front and rear cover) up through the gantry.
  - Follow up existing harness along vertical channel.
  - Route along the outside of the track.
  - Front connector to front cover.
  - Rear connector to rear cover.
3.) Connect connector J1of the adapter cable to J1 on the gantry bulkhead.
4.) Connect connector J1 of cable 2139577 to J1 on the adapter cable 2295232.
5.) Connect connectors to the front interference harness.
6.) Connect connectors to the rear interference harness.
7.) Feed long adapter cable end through horizontal cable channel to the table base.





2.3 Table Modifications

2.3.1 Overview

A new enhanced table control (ETC) board, Sanota board with remote tilt, and harness are installed in the table base.

WARNING MAKE SURE ALL LOCKOUT/TAGOUT PROCEDURES HAVE BEEN FOLLOWED BEFORE PERFORMING AN UPGRADE TO THESE SYSTEMS.

2.3.2 ETC Board

NOTICE Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist and to bare metal grounding point on the table before you continue.

The ETC board is located in the base of the table.

1.) Use a flat-blade screwdriver to loosen two screws on ETC cover.
2.) Remove ETC cover from the ETC housing.
3.) Tag and disconnect all connectors from the ETC board.
4.) Use a hex key to remove 14 screws that secure the ETC board and CT computer board to the ETC housing.

5.) Remove ETC board (with the ETC I/F board) and CT computer board as one from the ETC housing.

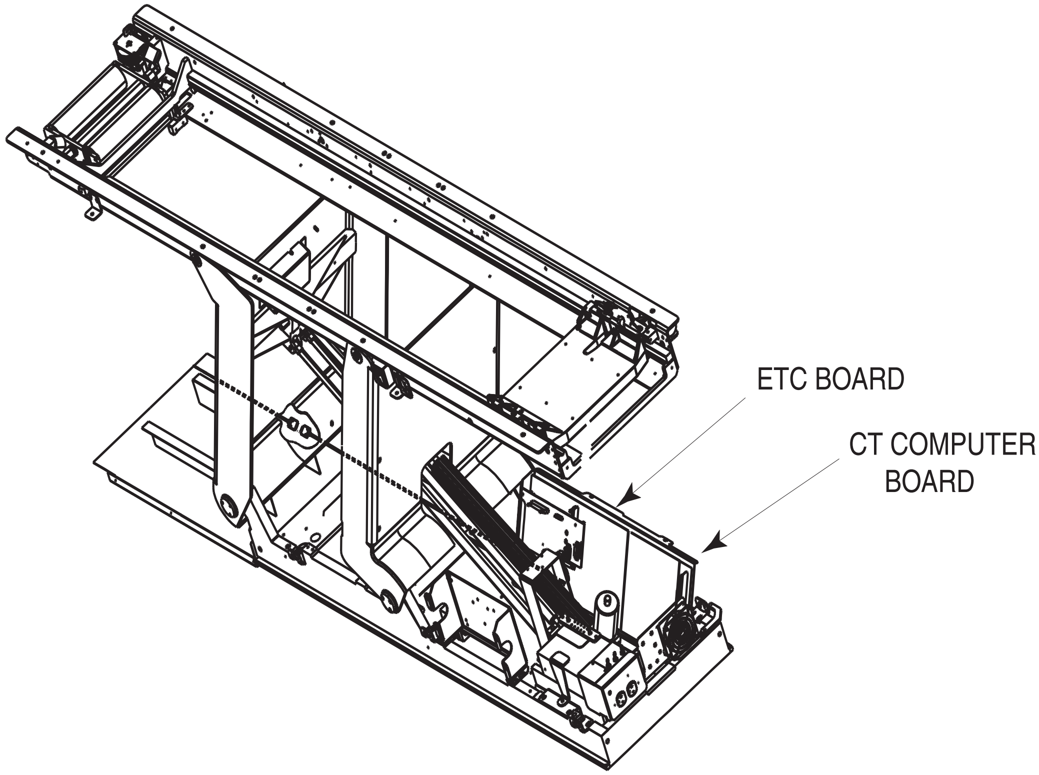


Figure 2-11 ETC Board and CT Computer Board on Table

6.) Disconnect ETC board from the CT computer board.

Note: Do not remove Sanota board with remote tilt or the ETC I/F board separately. Each will be removed with the ETC board.

7.) If present (CX/i only), remove Network card from the ETC board. Remove the nylon standoffs from the old ETC board, and save them to install on the new ETC board.

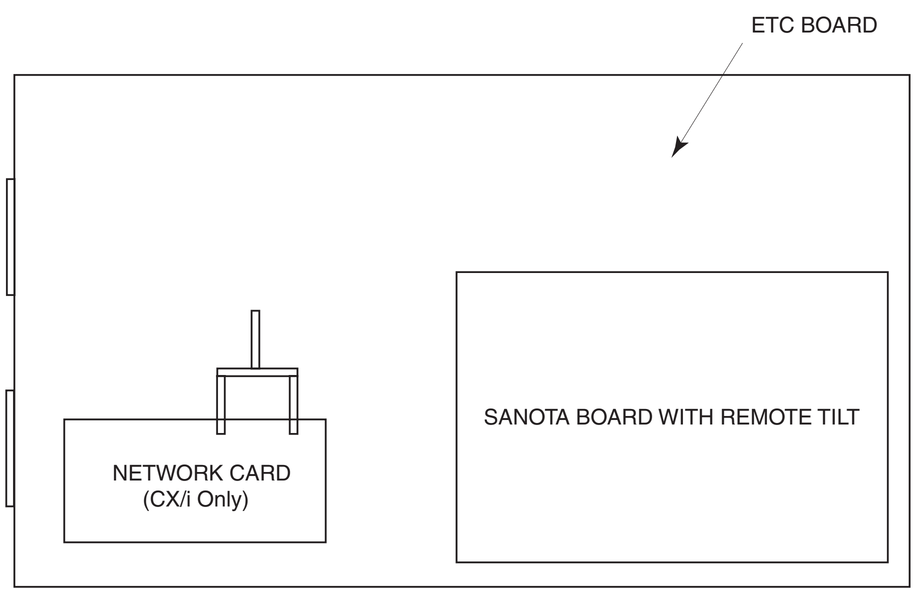


Figure 2-12 Location of Network Card on ECT Board

8.) If present (CX/i only), connect network card to the new ETC board 46-264368G3.
9.) Connect new ETC board 46-264368G3 to the CT computer board.
10.) Install new ETC board with CT computer board onto the ETC housing.
11.) Use a hex key to install 12 of the 14 screws (removed in step 4) securing the ETC board and CT computer board to the ETC housing. (The two unused screws will be used in step 14.)
12.) Cut tie-wrap from end of bundled cables with J6, J7, and J9 connectors.
13.) Connect J6 and J9 connectors to the new ETC board. (Connector J7 will be connected to the Sanota board with remote tilt according to step 19.)
14.) Add two brass standoffs 46-208984P17 to the ETC board.

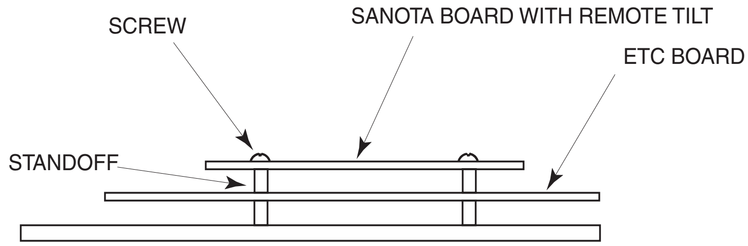


Figure 2-13 Mounted ETC Board and Sanota Board with Remote Tilt Board

15.) Install ribbon cable 2299221 to the following connectors:
- J4 on the ETC board
- J4 on the Sanota board with remote tilt according to step 19

16.) Install ribbon cable 2271254 to the following connectors:
- J5 on the ETC board
- J5 on the Sanota board with remote tilt according to step 19

17.) Install jumper cable 2271255 to the following connectors:
- J16 on the ETC board
- J3 on the Sanota board with remote tilt according to step 19

18.) Install new Sanota board with remote tilt 2290734 onto brass standoffs and secure with two screws 46-208560P68.
19.) Connect remaining connectors to the ETC board and Sanota board with remote tilt.

2.3.3 Add Harness

The wire harness provides two functions: completes the remote tilt option circuit, and provides cable disconnect during table service.

1.) Connect wire harness 2295233 to the following connectors:
- ETCI/F on adapter cable 2295230 or 2295232 from gantry
- J8 on the Sanota board with remote tilt 2990734

2.) Allowing for a service loop at the harness connectors, tie wrap wires harness (approximately every 6 to 8 inches) to table base.
3.) Route wire harness into ETC housing following other harnesses.
4.) Place area of wire harness with copper tape with others cables with copper tape and secure with tie wraps. (This will achieve a good shielded ground.)

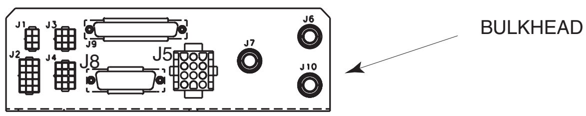
2.4 System Cables

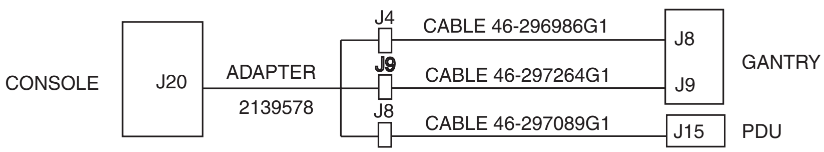
2.4.1 Overview

There are three console cable configurations on these systems.

2.4.1.1 Configuration #1

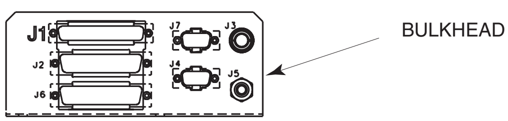
Adapter 2139578 connected to a bulkhead with two cables (46-296986G1 and 46-297264G1) going to the gantry, and one cable (46-297089G1) going to the PDU.

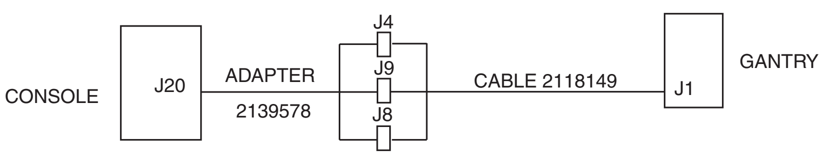




2.4.1.2 Configuration #2

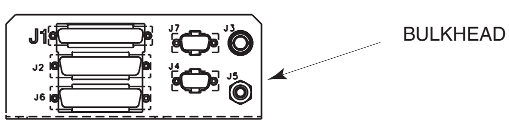
Adapter 2139578 connected to a bulkhead with one cable (2118149) going to the gantry.

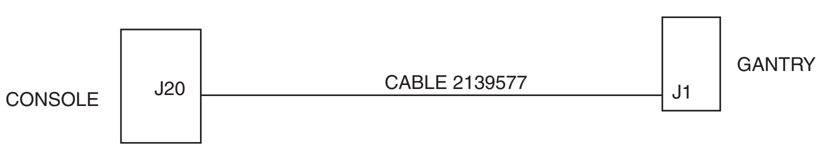




2.4.1.3 Configuration #3

No adapter - cable 2139577 going directly to the gantry.



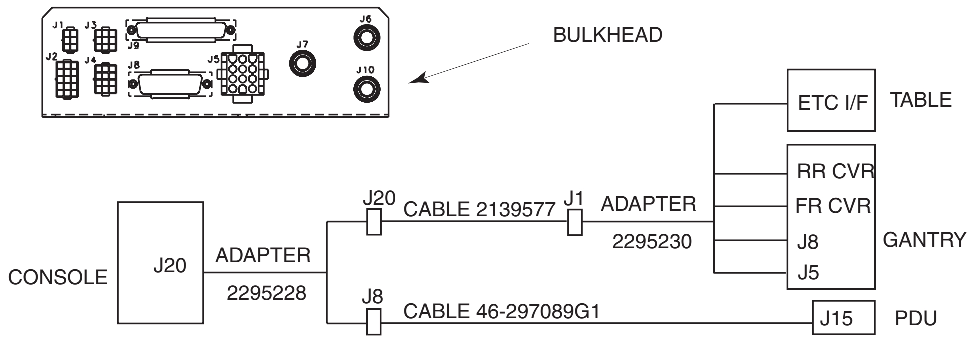


WARNING **MAKE SURE ALL LOCKOUT/TAGOUT PROCEDURES HAVE BEEN FOLLOWED BEFORE PERFORMING AN UPGRADE TO THESE SYSTEMS.**

NOTICE **Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist and to bare metal grounding point on the table before you continue.**

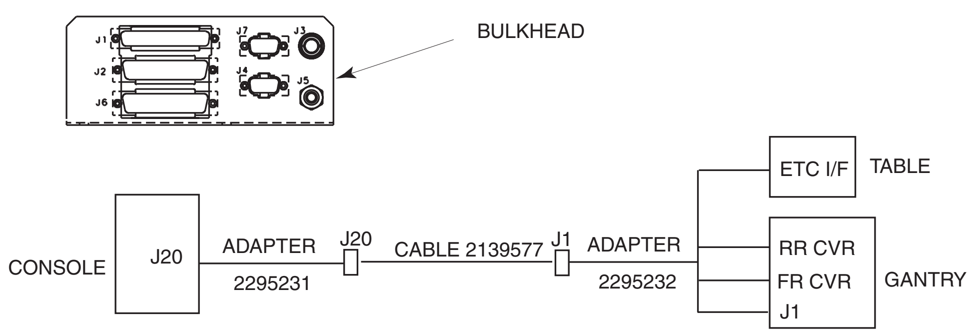
2.4.2 1st Configuration

1.) Disconnect adapter cable 2139578 from the following connectors:
    - J20 on the console
    - J4 on cable 46-296986G1
    - J8 on cable 46-297089G1
    - J9 on cable 46-297264G1
2.) Disconnect keyboard cable from J19.
3.) Remove cable 46-296986G1 from cable channel.
4.) Remove cable 46-297264G1 from cable channel.
5.) Install adapter cable 2295228 to J20 on the console after installing new board per Figure 2.5.1).
6.) Install cable 2139577 into cable channel with J20 connector end at the console.
7.) Connect cable 2139577 to the following connectors:
    - J20 to adapter cable 2295228
    - J1 to adapter cable 2295230 at the gantry
8.) Connect existing 46-297089G1 cable to J8 at adapter cable 2295228



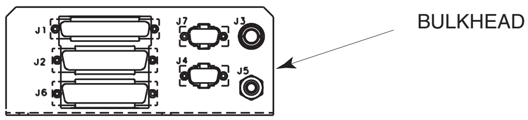
2.4.3 2nd Configuration

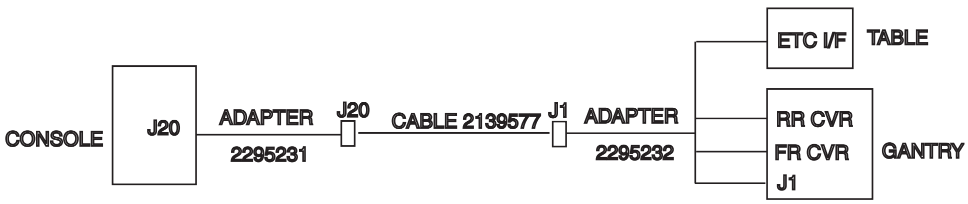
1.) Disconnect adapter cable 2139578 from J\20 on console.
2.) Disconnect keyboard cable from J19.
3.) Remove cable 46-2118149 and adapter 2139578 from cable channel.
4.) Install adapter cable 2295231 to J20 on the console (after installing new board per Figure 2.5.1).
5.) Install cable 2139577 into cable channel with J20 connector end at the console.
6.) Connect cable 2139577 to the following connectors:
    - J20 to adapter cable 2295231
    - J1 to adapter cable 2295232 at the gantry



2.4.4 3rd Configuration

1.) Disconnect cable 2139577 from J20 on console.
2.) Disconnect keyboard cable from J19.
3.) Install adapter cable 2295231 to J20 on the console (after installing new board per Figure 2.5.1).
4.) Connect cable 2139577 to the following connectors:
   - J20 to adapter cable 2295231
   - J1 to adapter cable 2295232 at the gantry





2.5 Console Modifications

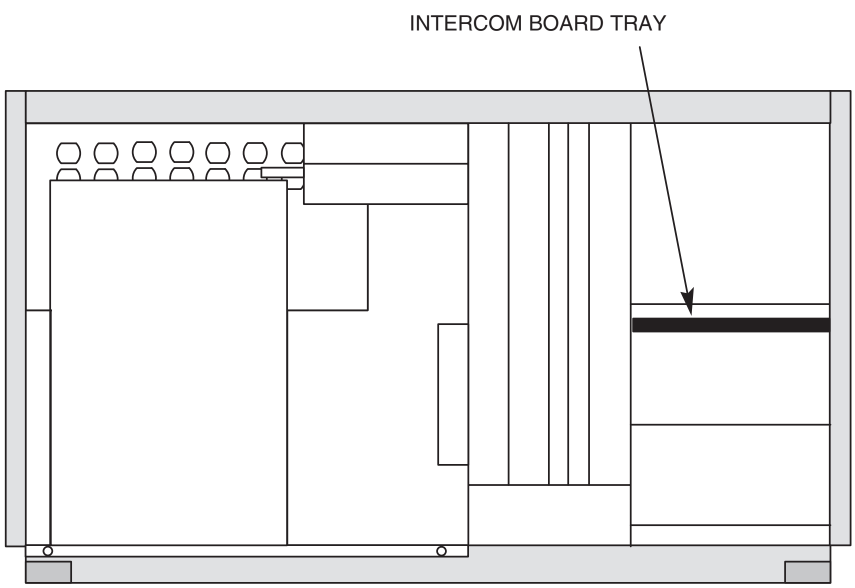
The Remote Tilt Upgrade includes a new intercom board, keyboard, and SCIM.

WARNING MAKE SURE ALL LOCKOUT/TAGOUT PROCEDURES HAVE BEEN FOLLOWED BEFORE PERFORMING AN UPGRADE TO THESE SYSTEMS.

NOTICE Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist and to bare metal grounding point on the table before you continue.

2.5.1 Intercom Board Configurations

1.) Loosen two screws that secure the intercom board tray to the console.



2.) Carefully pull the intercom board outward from the console until you see the connectors.
3.) Tag and disconnect power cord and connector from the intercom board.
4.) Remove intercom board tray from the console.
5.) Slide new intercom board tray (2169590-2) into the console.
6.) Connect connector and power cord to the intercom board.
7.) Push the intercom board into the console.
8.) Tighten two screws that secure the intercom board tray to the console.
9.) Connect adapters 2295228 or 2295231 to J20 on new intercom board tray.

2.5.2 SCIM and Keyboard

The new SCIM and keyboard replace the existing keyboard used with the Octane computer.

Note: Do not disconnect track ball from the console.

1.) Disconnect existing keyboard cable from the Octane backplane. This cable is no longer used and can be left in the bottom of the console.
2.) Remove keyboard from tabletop of the console.
3.) Place SCIM onto tabletop of the console.
4.) Place applicable overlay (with tilt) onto the SCIM.
5.) Connect SCIM cable to the SCIM.
6.) Connect SCIM cable into J19 connector in rear of the console.
7.) Route the keyboard cable under the SCIM and place onto SCIM plate.

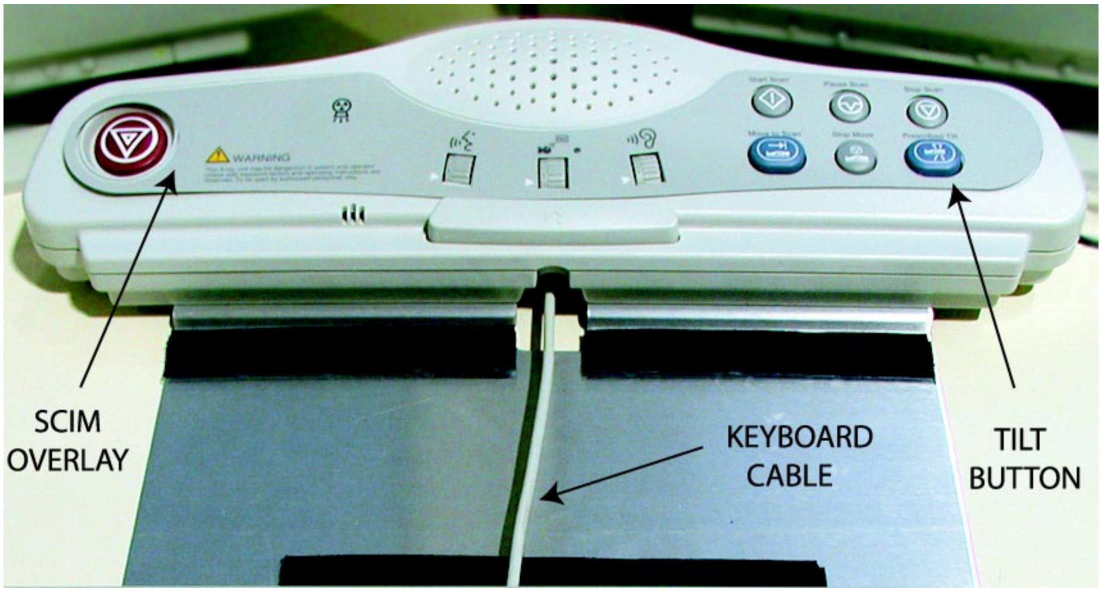


Figure 2-14 SCIM, showing keyboard cable routing

8.) Connect keyboard cable to Octane backplane.
9.) Place applicable overlay onto the keyboard.

2.6 Software

The software revision must be at least 6.3 (CT/i) or 1.4 (QX/i). Load software now if necessary.

2.7 Rating Plates

1.) Install rating plate for remote tilt upgrade next to existing rating plates on the console.
2.) Install rating plate for remote tilt upgrade next to existing rating plates on the table.
3.) Install rating plate for remote tilt upgrade next to existing rating plates on the gantry.

2.8 Install Covers

To gain access to internal components of the gantry, table base, and console, covers will need to be removed: Once the upgrade is complete, covers are installed in the reverse order.

2.8.1 Console

Install front cover from the console. Refer to *System Installation Manual*, 2152926-100.

2.8.2 Table

1.) Install all covers between the table base and gantry base. Refer to *System Installation Manual*, 2152926-100.
2.) Install right base cover from the table base. Refer to *System Installation Manual*, 2152926-100.

2.8.3 Gantry

1.) Install top covers from the gantry. Refer to *System Installation Manual*, 2152926-100.
2.) Install side covers from the gantry. Refer to *System Installation Manual*, 2152926-100.

Section 3.0
Customer Handover

Once all components of the upgrade are installed, system hardware checked, software has been verified, and the site cleaned, the system can be handed over to the customer.

3.1 Mechanical Interface Checks

3.1.1 Check SmartView Option

Check that SmartView Option is working properly, if applicable. Refer to SmartView section of the Operators Manual for procedure.

3.1.2 Check Remote Tilt Option

Check that Remote Tilt Option is working properly. Refer to Operators Manual for procedure.

3.1.3 Check Sensor Functions

3.1.3.1 Front Sensor

1.) Tilt gantry and touch front sensor. Tilt should stop and warning light should come on.
2.) Tilt gantry until light goes off.
    - If light goes OFF after a short delay, sensor is working properly.
    - If light does not go off after approximately 5 degrees, sensor is not working properly.
3.) Repeat last step touching front sensor in three various locations.

3.1.3.2 Rear Sensor

1.) Tilt gantry and touch rear sensor. Tilt should stop and warning light should come on.
2.) Tilt gantry until warning light goes off.
    - If light goes OFF after a short delay (approx. 5 seconds), sensor is working properly.
    - If light does not go off after approximately 5 degrees, sensor is not working properly.
3.) Repeat last step touching rear sensor in five various locations

3.1.4 Table Checks

1.) Check that table moves prior to a scan.
    Move Table from current position.
    (Result should be table moving from one location to another.)
2.) Stop table during positioning of table.
    Move table to another position.
3.) Move table up and down.
    - Move table up.
    - Move table down.
      (Result should be table movement in both up and down positions.)

3.1.5 Keyboard and SCIM Checks

1.) Check E-stop button.
    - Tilt the gantry from the console.

- Press the E-stop button on the SCIM.
  (Result should be gantry stops tilt motion.)

2.) Check intercom functions.
- Speak into intercom at console
  (Result should be clear and proper audible volume at gantry)
- Speak into microphones in gantry.Result should be playback message.
  (Result should be clear and proper audible volume at console.)

3.) Check voice record and playback.
- Record message.
- Play recorded message.
  (Result should be clear and proper audible volume.)

4.) Check auto voice message.
- Select desired message.
- Play auto message.
  (Result should be clear and proper audible volume at gantry.)

3.1.6 Scan Checks

1.) Check start scan function.
- Select scan type.
- Start scan by initiating a scan.
  (Result should be scan will start.)

2.) Check pause of a scan function.
- Select a scan.
- Start scan by initiating a scan.
- Pause scan.
  (Result should be the scan will start and then pause.)

3.) Check stop of a scan function.
- Select a scan.
- Start scan by initiating a scan.
- Stop scan.
  (Result should be stopping of a scan in process.)

3.2 Review Upgrade with Customer

Review the following Remote Tilt Upgrade features with the customer.
- Explain how the Tilt feature can be performed from the console.
- Tilt gantry forward and activating the front sensor.
- Tilt gantry backward and activating the rear sensor.
- Perform scans typical per site requirements.
- Make sure customer is comfortable with new features before leaving site.

Section 4.0
Theory

The current LightSpeed systems control the tilt function through the gantry control panel only. The Remote Tilt Upgrade adds the ability to control the tilt function from both the gantry control panel and the console. Functionally, tilt has not changed with this upgrade.

Refer to Figure 4-1. Upon depression of the remote tilt enable button on the SCIM board, a signal is passed from the console through the gantry bulkhead to the Enhanced Sanota board. The Enhanced Sanota board checks the gantry interference switch conditions continuously as it passes the tilt command to the ETC board registers. Based on the prescription, the gantry is positioned to the prescribed tilt location

The gantry mounted controls and hand held controls also pass their signals through the Enhanced Sanota board to the ETC for positioning table and gantry.

The command to drive the tilt amplifier is sent from the ETC board through the Enhanced Sanota board to the tilt amplifier. This causes the tilt motor/pump to move the gantry in the desired direction. Once the proper angle is achieved, this signal is removed.

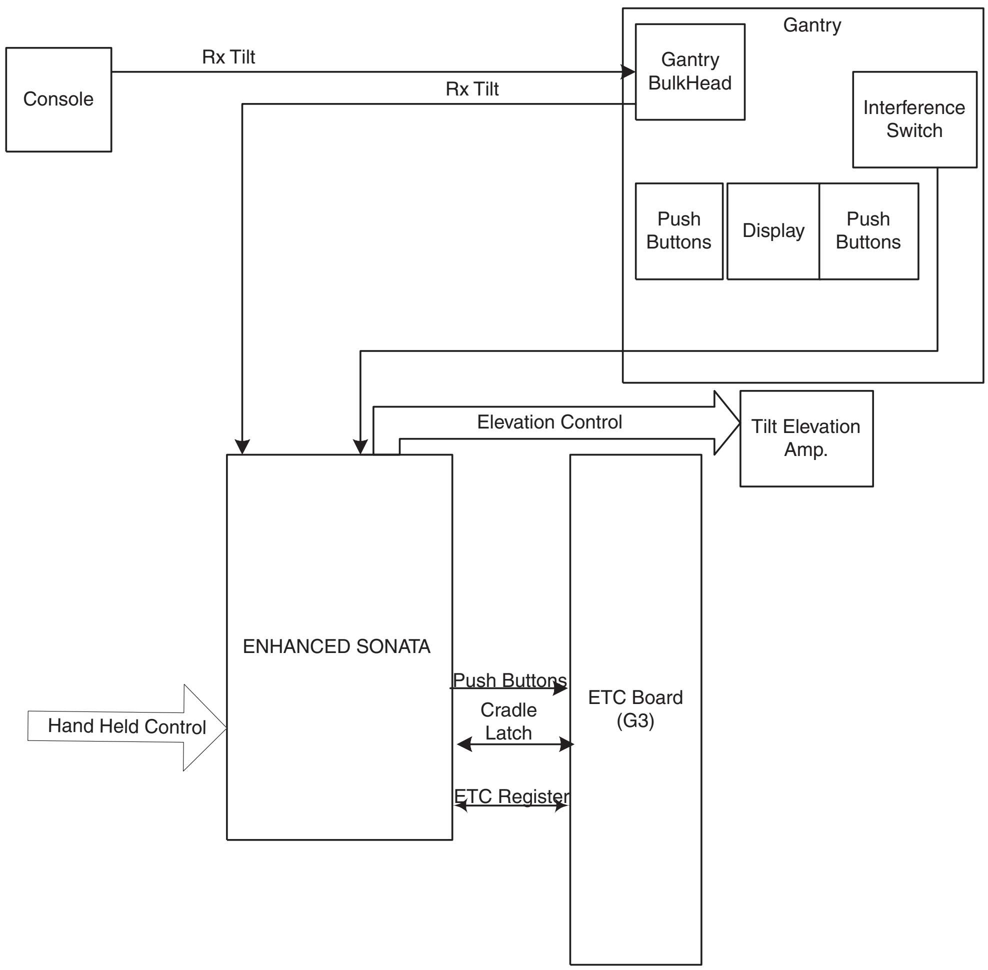


Figure 4-1 System Block Diagram

The Enhanced Sanota board is designed to carry the signals required for the remote tilt upgrade. The additional signals include changing the push button interface, allowing for remote tilt from the console and interference sensors on the gantry.

4.1 Interference Detection

Sensors are mounted on the front and rear of the gantry. When the sensor is not pressed the impedance is 20 kOhms. When the sensor is pressed the resistance becomes near 0 Ohms.

For both front and rear sensors, the signal will be decoded on the board to determine if the maximum impedance of the sensor, 40 kOhms, has been exceeded creating a fault condition or if the sensor is in a short circuit condition, less than 20 kOhms, indicative of interference. This is done for both front and rear sensors.

This information is passed through register FFAE22 on the ETC board. If there is interference or a fault condition both elevation and tilt will be disabled. A green LED (DS7 and/or DS8) will indicate that the sensor is not short-circuited. A yellow LED (DS7 and/or DS8) will indicate that the sensor has more than 40 kOhms of resistance and has failed.

4.2 Tilt and Elevation Interface

The Enhanced Sanota board intercepts the signals that go to the tilt elevation amplifier from the ETC board. The signals required for generating tilt (forward and backward) will be decoded and sent to the tilt elevation Amp.

The signal enabling elevation is intercepted and another enable is created so that the interface board can disable elevation if the interference sensor is faulted- or interference is detected.

4.3 Timing Circuit

A 555 Timer disables push buttons from exiting the board if the four Rtilt, 2 high and 2 low, signals are not the same for more than 0.99 seconds. When the Push Button from the console is pressed, it sends a signal to the ETC board. This is a hardware failure monitor, similar to the stuck button indicators.

4.4 Remote Tilt

Refer to Figure 4-2, two lines must be active to initiate remote tilt. When both lines are active the Altera generates a signal to the ETC board through register FFAE22. The ETC board must check for stuck button conditions.

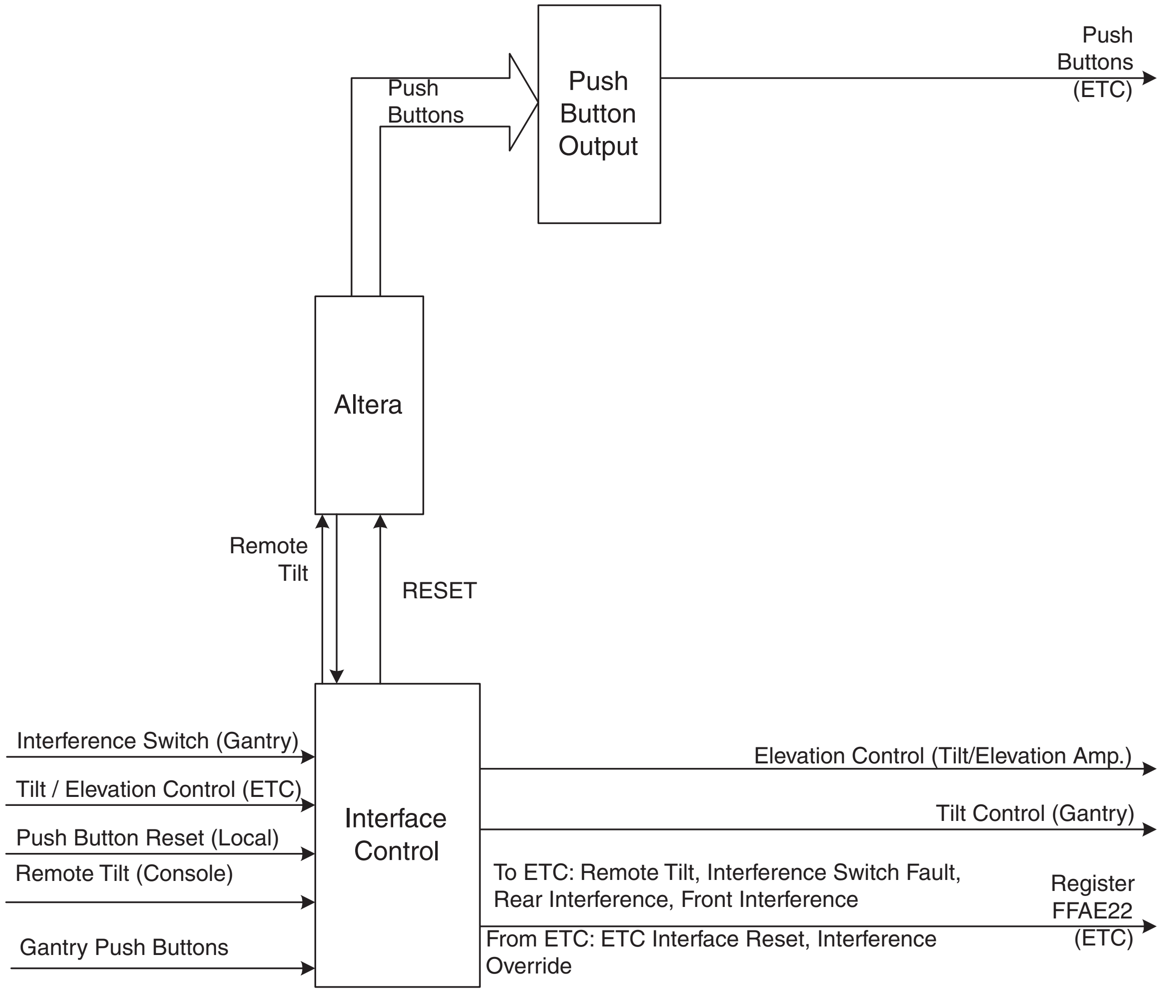


Figure 4-2 Top Level Diagram

Section 5.0
Servicing

5.1 Corrective Maintenance

There are seven Field Replaceable Units (FRU) to repair in the Remote Tilt Upgrade.

1.) Enhanced Table Control (ETC) board
2.) Enhanced Sanota board with remote tilt
3.) Front sensor
4.) Rear sensor
5.) Keyboard
6.) SCIM
7.) Intercom/Presecribed Tilt Board

5.1.1 Detect

Perform Remote Tilt functions. Refer to the Operators Manual. Once all functions have been verified to be operating properly, hand over to the customer.

5.1.2 Diagnose

5.1.2.1 ETC Board

Perform Tilt Diagnostics. Refer to the Service Manual.

5.1.2.2 Enhanced Sanota Board

LEDs

The Enhanced Sanota Board has fourteen LEDs to aid in troubleshooting remote tilt.

LED Number	Name	Status
DS1	Remote Tilt Hardline 2	Signal from console. Will extinguish upon receipt of a console signal.
DS2	Remote Tilt Hardline 1	Signal from console. Will extinguish upon receipt of a console signal.
DS3	Vcc Power On.	Logic +5 Volt supply active when lit.
DS4	Tilt LED	Tilt or Remote Tilt Button is depressed when lit.
DS5	Elevation LED	Elevation button is depressed when lit.
DS6	Rear Sensor Touch	The gantry rear touch sensor is depressed when lit.

Table 5-1 Enhanced Sanota Board LEDs

LED Number	Name	Status
DS7	Front Sensor Fault	The gantry front touch sensor has failed when yellow.
DS8	Rear Sensor Fault	The rear gantry touch sensor has failed when yellow.
DS9	Front Sensor Touch	The gantry front touch sensor is depressed when lit.
DS10	(Smartview) Relay 1	Used during Smartview for cradle adjustments. Lit indicates relay is energized.
DS11	(Smartview) Relay 2	Used during Smartview for cradle adjustments. Lit indicates relay is energized.
DS12	24 V Power	24 Volt power supply is active when lit.
DS13	(Smartview) Relay 3	Used during Smartview for cradle adjustments. Lit indicates relay is energized.
DS14	(Smartview) Relay 4	Used during Smartview for cradle adjustments. Lit indicates relay is energized.

Table 5-1 Enhanced Sanota Board LEDs (Continued)

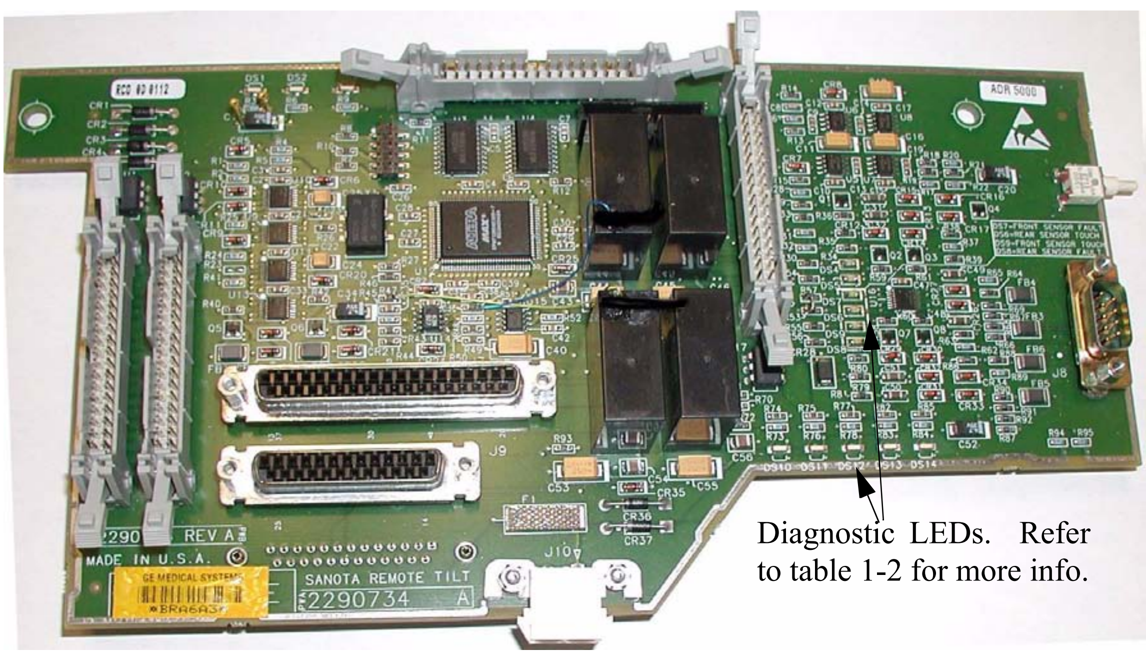


Figure 5-1 Enhanced Sanota Board

Connector Pin Outs

Pin Number	Signal
1	RTILT_L1_H
6	RTILT_L1_L
2	RTILT_L2_H
7	RTILT_L2_L
4	RCVR_TCH1
9	RCVR_TCH2
3	FCVR_TCH1
8	FCVR_TCH2

Table 5-2 J14 (from Console) Sub D

Pin Number	Signal
1	ETCDATA0
2	ETCDATA1
3	ETCDATA2
4	ETCDATA3
5	ETCDATA4
6	ETCDATA5
7	ETCDATA6
8	ETCDATA7
9	ETC_WR*
10	ETC_RD*

Table 5-3 J3 (ETC VME FFAE22)

Pin	Signal
1	TILT_FWD_PB*
2	TILT_FWD_PB*
3	LGND
4	LGND
5	ALIGN_LITE_PB*
6	ALIGN_LITE_PB*
7	ADV_PB*
8	ADV_PB*
9	LGND
10	LGND
11	INT_PB*
1‘2	INT_PB*
13	LGND
14	LGND
15	LGND
16	ELEV_DOWN_PB*
17	FAST_PB*
18	LGND
19	LGND
20	ELEV_UP_PB*
21	CRADLE_IN_PB*
22	CRADLE_OUT_PB*
23	LGND
24	LGND
25	EXT_PB*
26	LGND
27	LGND
28	LGND
29	LGND
30	LGND
31	TILT_BACK_PB*
32	TILT_BACK_PB*
33	LGND
34	LGND

Table 5-4 J5 Sub D

Pin	Signal		Pin	Signal
1	NC		1	NC
2	ELEV_EN*		2	ELEV_OUT_EN*
3	NC		3	NC
4	NC		4	NC
5	NC		5	NC
6	SPARE_DRIVER_1*		6	SPARE_DRIVER_1*
7	NC		7	NC
8	SPARE_DRIVER_2*		8	SPARE_DRIVER_2*
9	TILT_EN*		9	TILT_EN*
10	NC		10	NC
11	E/T_AMP_EN		11	E/T_AMP_EN
1‘2	+24V_ETC_RTN		1‘2	+24V_ETC_RTN
13	+24V_ETC		13	+24V_ETC
14	+24V_ETC_RTN		14	+24V_ETC_RTN
15	NC		15	NC
16	NC		16	NC
17	E/T_CW-TOP		17	E/T_CW-TOP
18	E/T_CW-TOP_RTN		18	E/T_CW-TOP_RTN
19	E/T_CCW-TOP		19	E/T_CCW-TOP
20	E/T_CCW-TOP_RTN		20	E/T_CCW-TOP_RTN
21	E/T_CW-BOTTOM		21	E/T_CW-BOTTOM
22	E/T_CW-BOTTOM_RTN		22	E/T_CW-BOTTOM_RTN
23	E/T_CCW-BOTTOM		23	E/T_CCW-BOTTOM
24	E/T_CVW-BOTTOM_RTN		24	E/T_CVW-BOTTOM_RTN
25	E/T_PULSE-IL*		25	E/T_PULSE-IL*
26	E/T_PULSE-IL_RTN		26	E/T_PULSE-IL_RTN
27	E/T_V-MOTOR		27	E/T_V-MOTOR
28	E/T_V-MOTOR_RTN		28	E/T_V-MOTOR_RTN
29	NC		29	NC
30	NC		30	NC
31	+15V_ETC		31	+15V_ETC
32	-15V_ETC		32	-15V_ETC
33	+15V_ETC		33	+15V_ETC
34	-15V_ETC		34	-15V_ETC

Table 5-5 J4 (to ETC J4) **J24 (to Tilt/Elevation Amplifier)**

5.1.2.3 Sensor - Front /Rear

Sensor is not functioning properly.

- Using an Ohm meter (between pins 1 and 2), check for open circuit.
  * Normal circuit should read 20K +200/-0 Ohms
  * Press on sensor should result in a reading of less than 100 Ohms

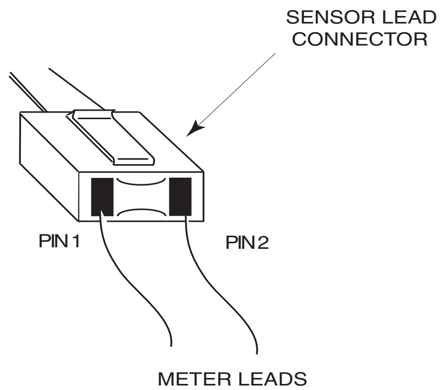


Figure 5-2 Sensor Connector Measurement Locations

- Replace sensor if not operating properly.

5.1.2.4 Keyboard

Perform Keyboard Diagnostics. Refer to the Service Manual.

5.1.2.5 SCIM

Perform SCIM Diagnostics. Refer to the Service Manual.

5.1.3 Replacement Instructions

5.1.3.1 ETC Board

Perform ETC Board Replacement procedure. Refer to Service Manual.

5.1.3.2 Enhanced Sanota Board with Remote Tilt

Required tools

- Small Flat-blade screwdriver
- Large Flat-blade Screwdriver
- ESD Wristband

NOTICE **Prevent permanent damage to static sensitive boards by attaching the anti static wrist strap to your wrist and to a bare metal grounding point on the table before you continue.**

1.) If possible, raise table to maximum elevation.
2.) Remove table base covers.
3.) Power off the table by flipping the three switches opposite the ETC assembly.
4.) Where installed, use a large flat-blade screwdriver to release the two screws holding the ETC assembly cover.
5.) Use a large flat-blade screwdriver to remove the screw on the floor of the table control area that allows the assembly to pivot.

6.) Pivot the assembly.
7.) Disconnect all connections to the Enhanced Sanota board.
8.) Remove two screws securing the Enhanced Sanota board to the ETC with small flat-blade screwdriver.
9.) Reverse procedure to reinstall.
10.) Verify operation of the Remote Tilt functions. Refer to Section 3.0.

5.1.3.3 Sensors - Front/Rear

WARNING MAKE SURE ALL LOCKOUT/TAGOUT PROCEDURES HAVE BEEN FOLLOWED BEFORE PERFORMING AN UPGRADE TO THESE SYSTEMS.

1.) Turn off power to the gantry.
2.) Open applicable front or rear cover.
3.) Disconnect sensor lead.
4.) Remove nuts with lockwashers and mounting screws from sensor.
5.) Remove sensor from cover.
6.) Remove all RTV from cover.
7.) Install sensor. Refer to section 2.2.1.1 or 2.2.2.1.

5.1.3.4 Keyboard

Perform Keyboard Replace procedure. Refer to Service Manual.

5.1.3.5 SCIM

Perform SCIM Replace procedure. Refer to Service Manual.

5.1.4 Verification

Verify operation of the Remote Tilt functions. Refer to Section 3.0.

5.2 Periodic Maintenance

No periodic maintenance is required for the Remote Tilt function.

Section 6.0
Sanota Board w/ Remote Tilt - 2290734

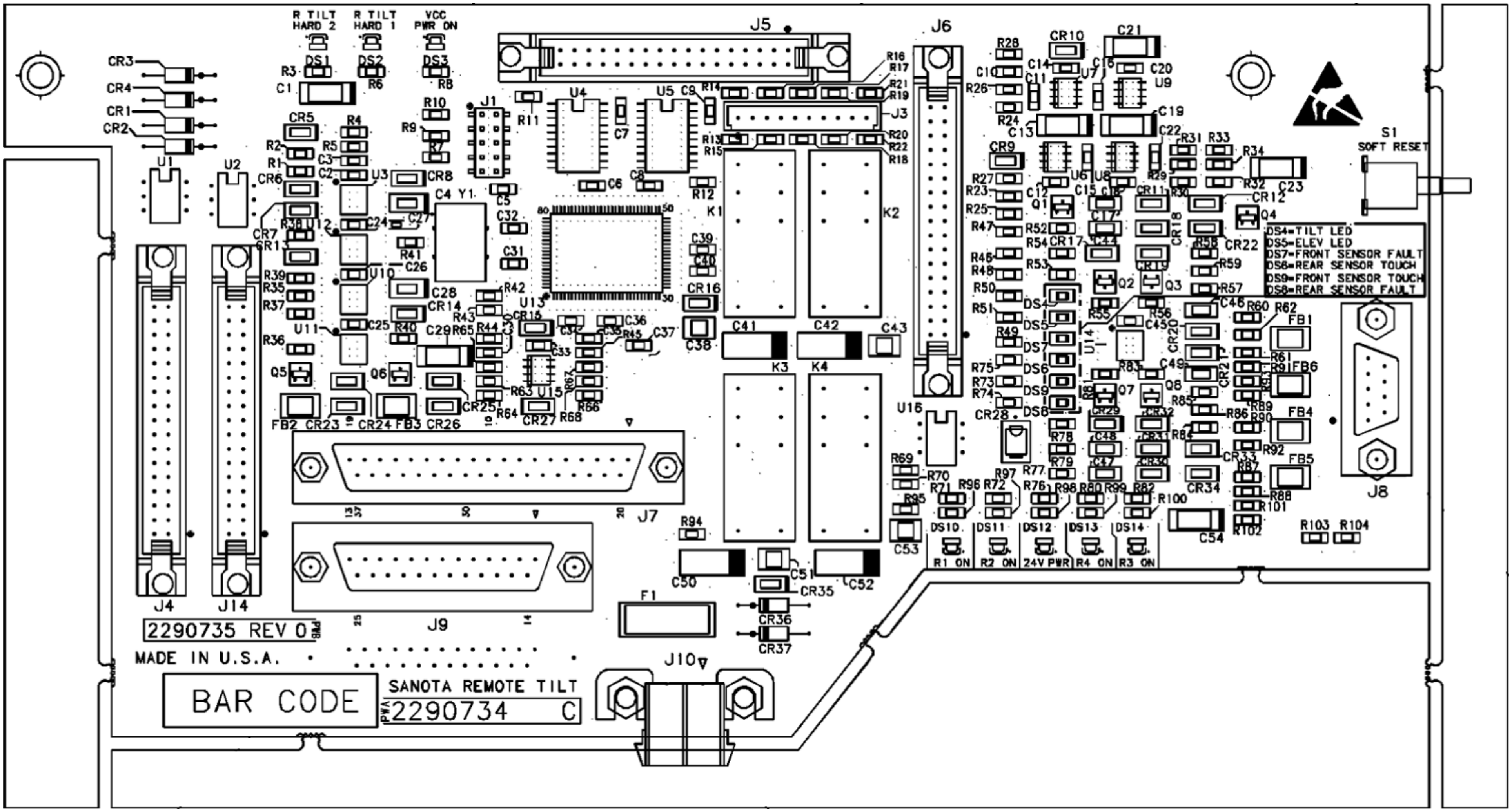


Figure 6-1 Sanota Board (with Remote Tilt) Layout (Sheet 1, Rev 0)

2290734SCH

NOTES:

1. THE GROUP NUMBER AND GROUP REVISION FOR THIS PWA IS 2290734 C
2. THIS SCHEMATIC MEETS THE SPECIFICATIONS OF GE SCHEMATIC STANDARDS 46-017486.
3. UNLESS OTHERWISE SPECIFIED:
   RESISTORS ARE IN OHMS, 0.100W, 1%.
   CAPACITORS ARE IN FARADS.
   INDUCTORS ARE IN HENRIES.
4. ACTIVE LOW SIGNALS ARE INDICATED WITH "*" AFTER THE SIGNAL NAME

SHEET_NAME	SHT NUM	REV
SONATA W/ RTILT	2	0
SONATA/TCH_SENS_IF	3	0
SONATA/TCH_SENS_IF	4	0
SONATA/CNSL_IF	5	0
SONATA/PB_IN	6	0
SONATA/TILT_CNTL	7	0
SONATA/TILT_CNTL	8	0
SONATA W/RTILT	9	0
SONATA/IHHC_RELAY	10	0
SONATA /INT_BLK_DIA	11	0

BLOCK PATHNAME $GEMS_CT_PAT_POS/2290734_ASM — BLOCK SHT 1 OF 9

REV C PCN N/A DESCRIPTION CHANGED REF DES; J4 TO J14, J5 TO J4, J3 TO J5, J2 TO J3 MADE BY BARB MERTES DATE 06/25/01 APPR BY DATE	MADE Michael Peete DATE 30/JAN/01 CHECK DATE	GE MEDICAL SYSTEMS MILWAUKEE, WI USA	DESIGN TITLE SONATA W/RTILT FIRST MADE FOR WARP 3
REV PCN DESCRIPTION MADE BY DATE APPR BY DATE	APPR DATE OTHER DATE		DRAWING NO. 2290734SCH — REV 0 — SIZE B — SHT/CONT. ON 1/ 2

Figure 6-2 2290734, Sheet 1, Rev 0

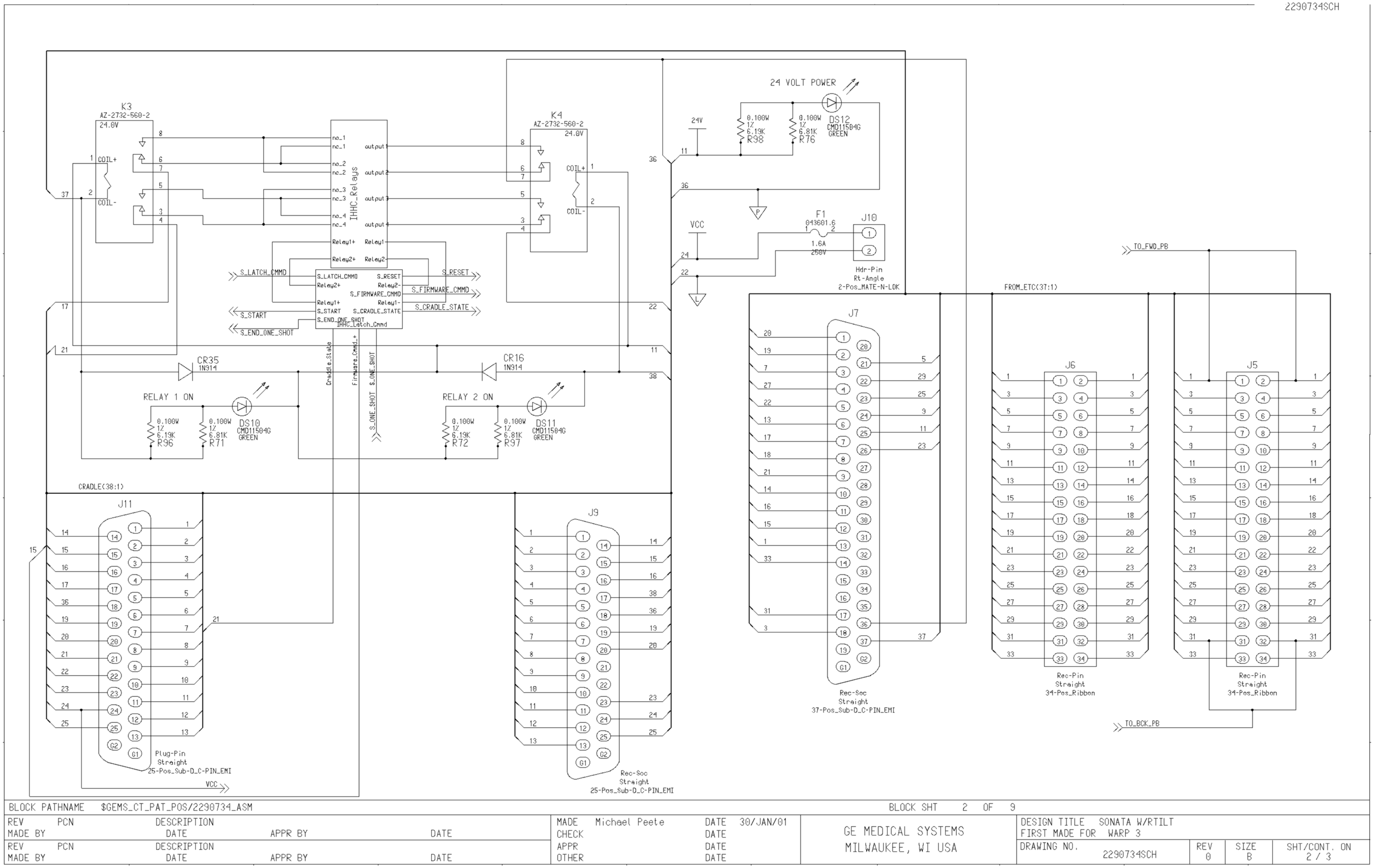


Figure 6-3 2290734, Sheet 2, Rev 0

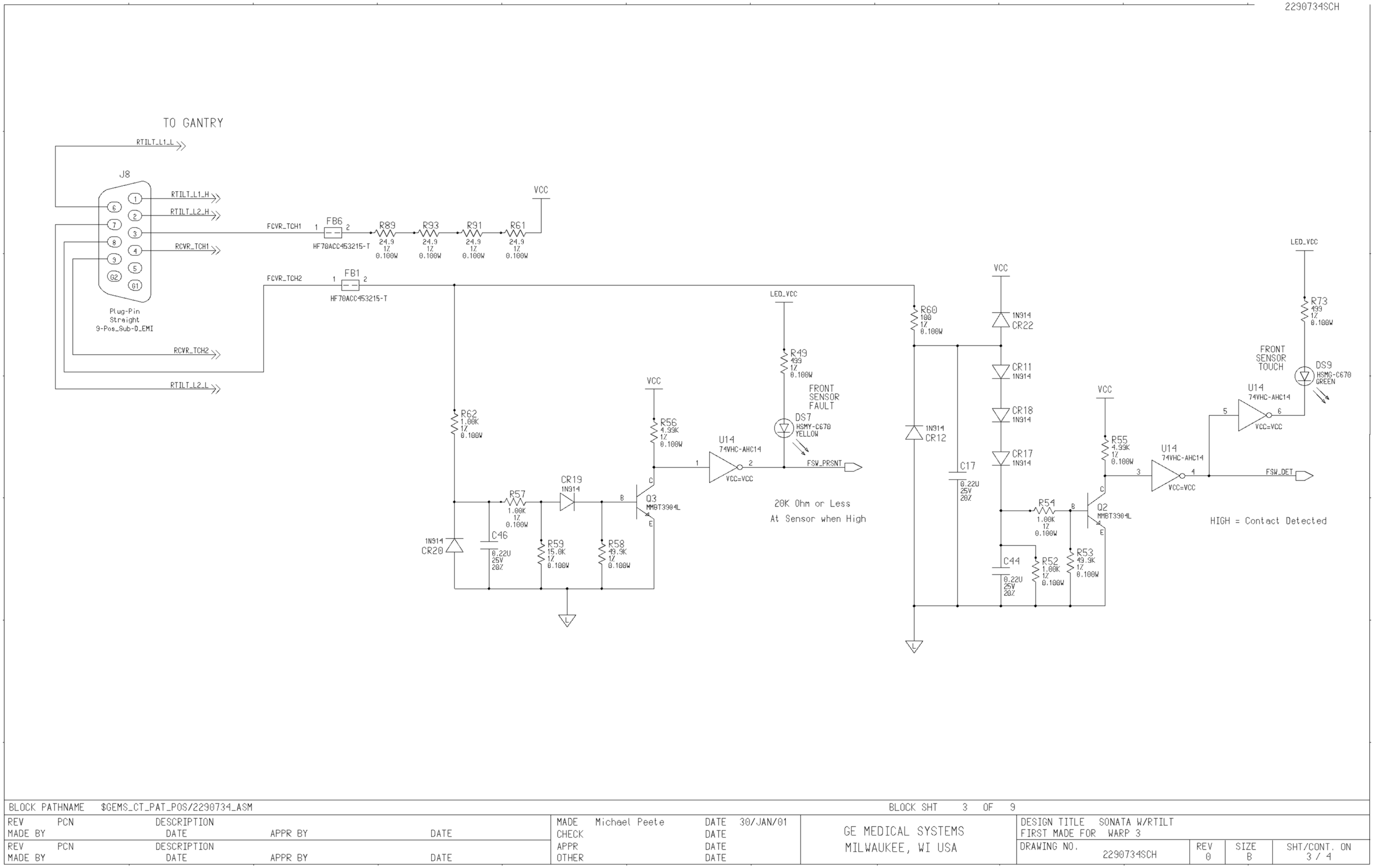


Figure 6-4 2290734, Sheet 3, Rev 0

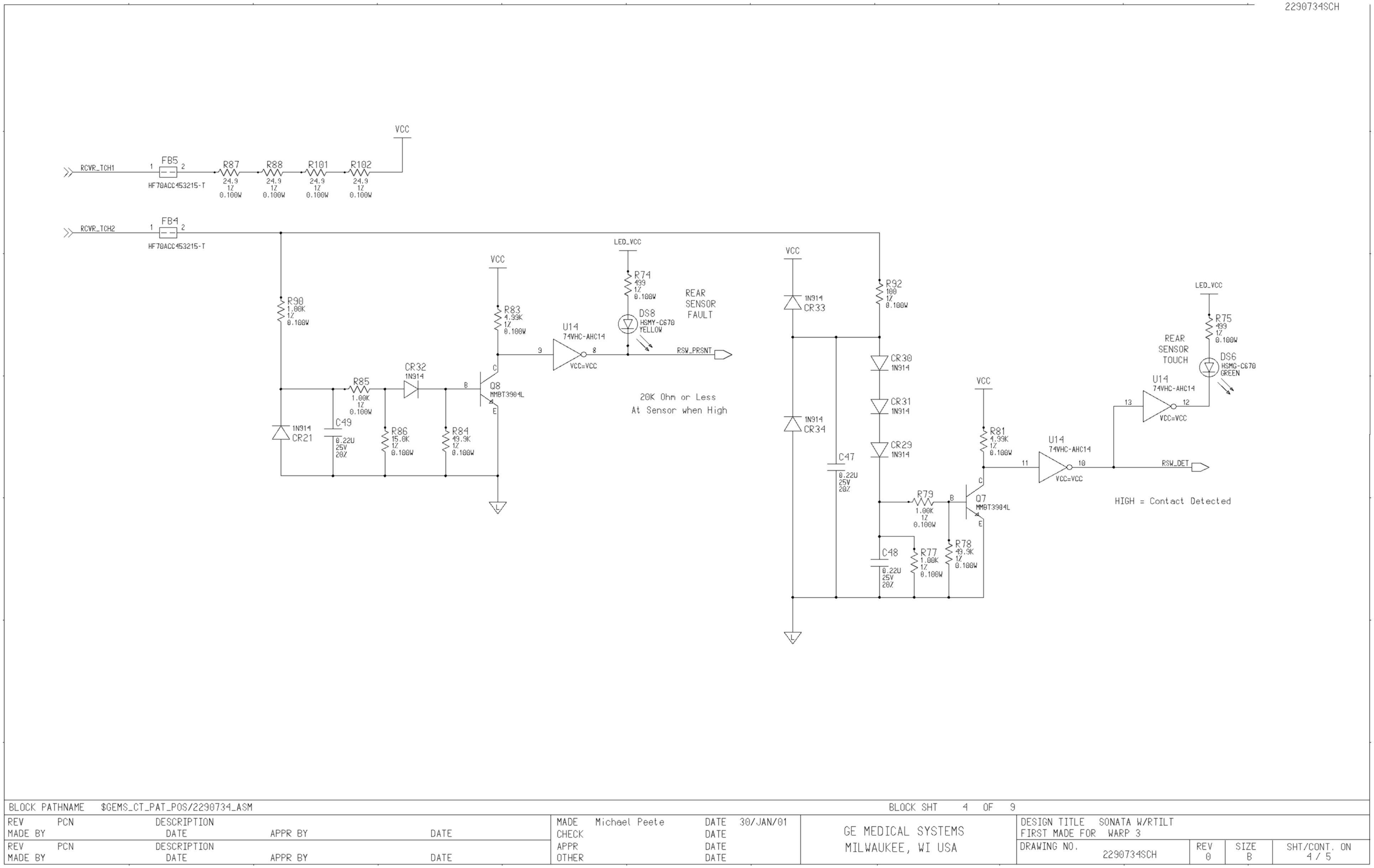


Figure 6-5 2290734, Sheet 4, Rev 0

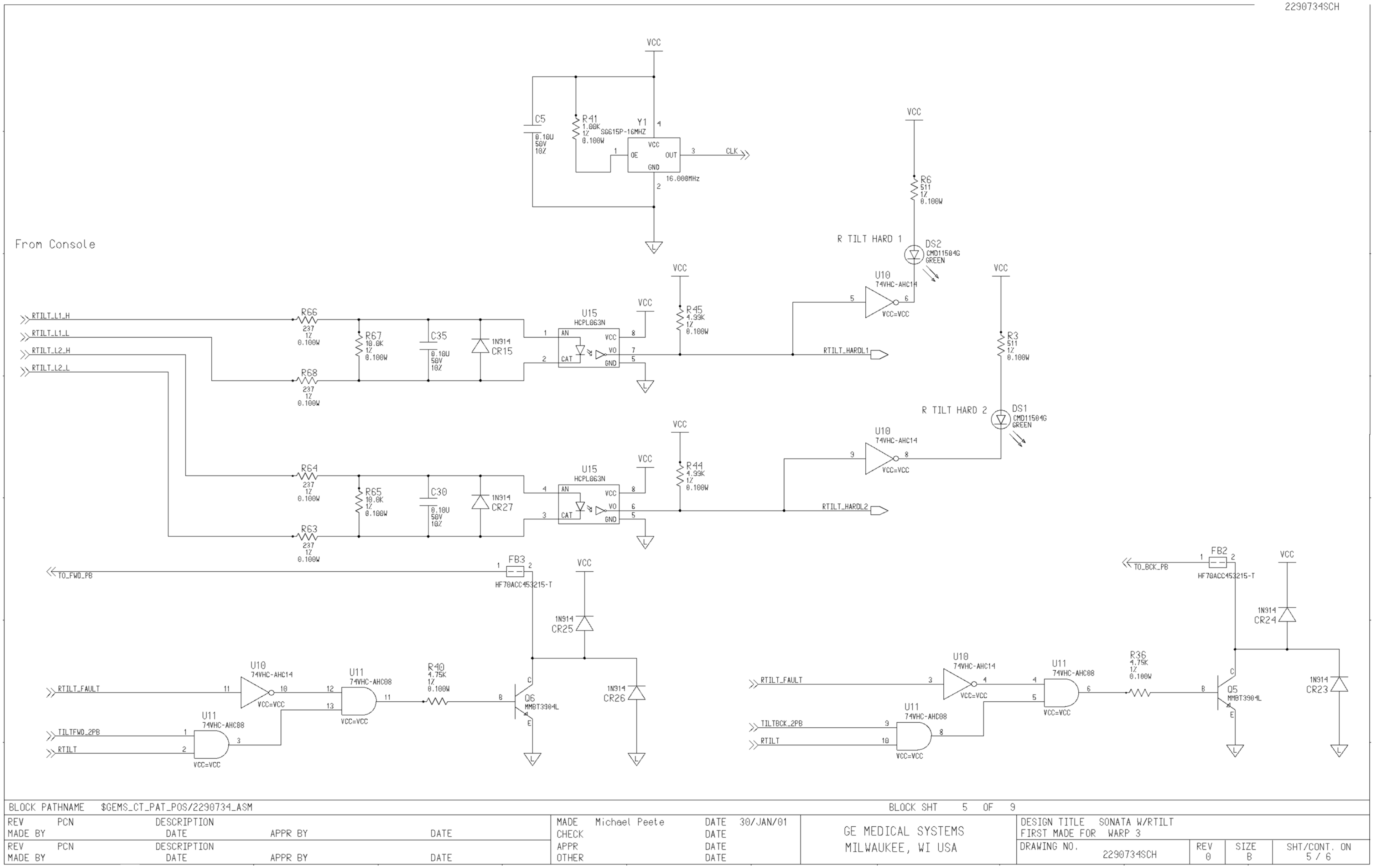


Figure 6-6 2290734, Sheet 5, Rev 0

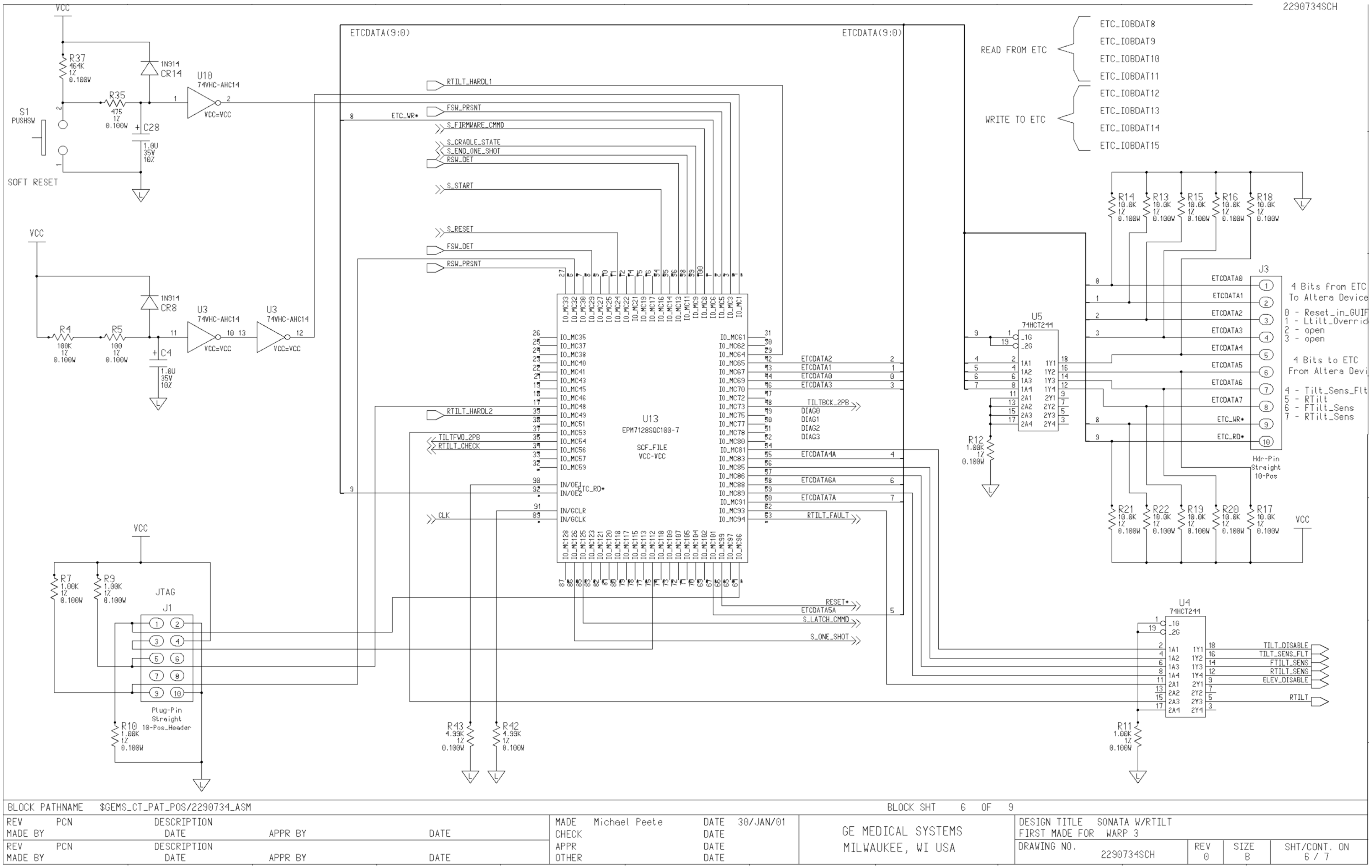


Figure 6-7 2290734, Sheet 6, Rev 0

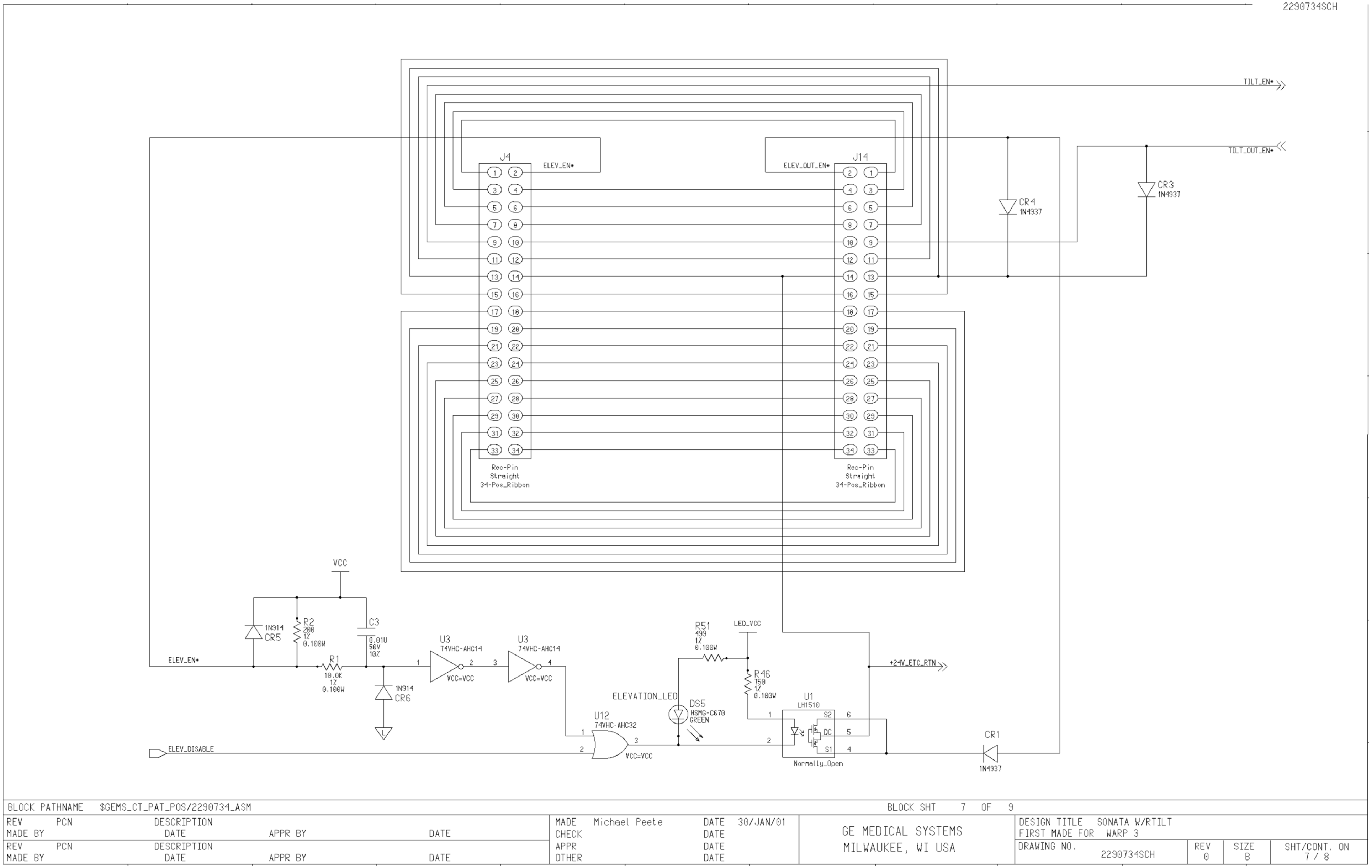
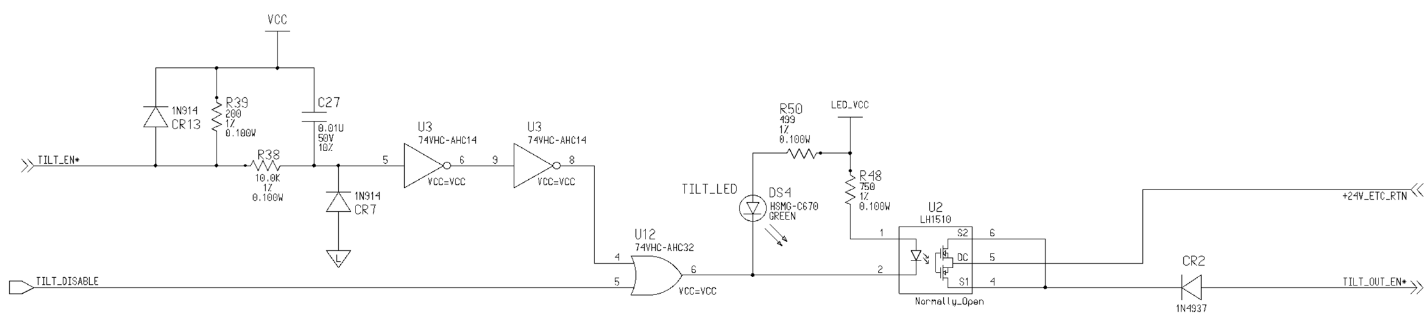


Figure 6-8 2290734, Sheet 7, Rev 0

2290734SCH



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REV PCN DESCRIPTION MADE BY DATE APPR BY DATE	MADE Michael Peete	DATE 30/JAN/01	GE MEDICAL SYSTEMS MILWAUKEE, WI USA	DESIGN TITLE SONATA W/RTILT FIRST MADE FOR WARP 3	
	CHECK	DATE			
REV PCN DESCRIPTION MADE BY DATE APPR BY DATE	APPR	DATE		DRAWING NO. 2290734SCH	REV 0 · SIZE B · SHT/CONT. ON 8 / 9
	OTHER	DATE			

Figure 6-9 2290734, Sheet 8, Rev 0

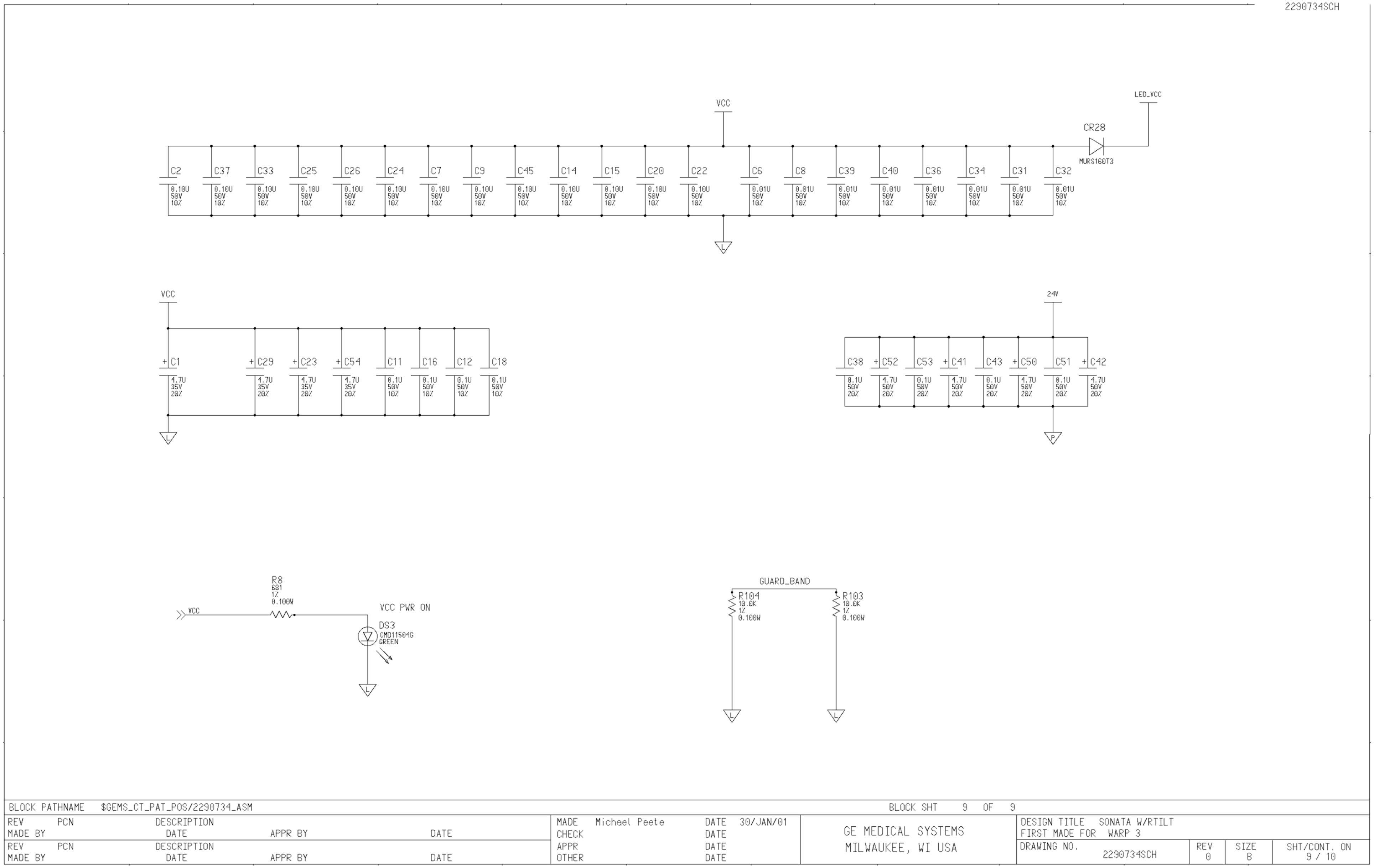
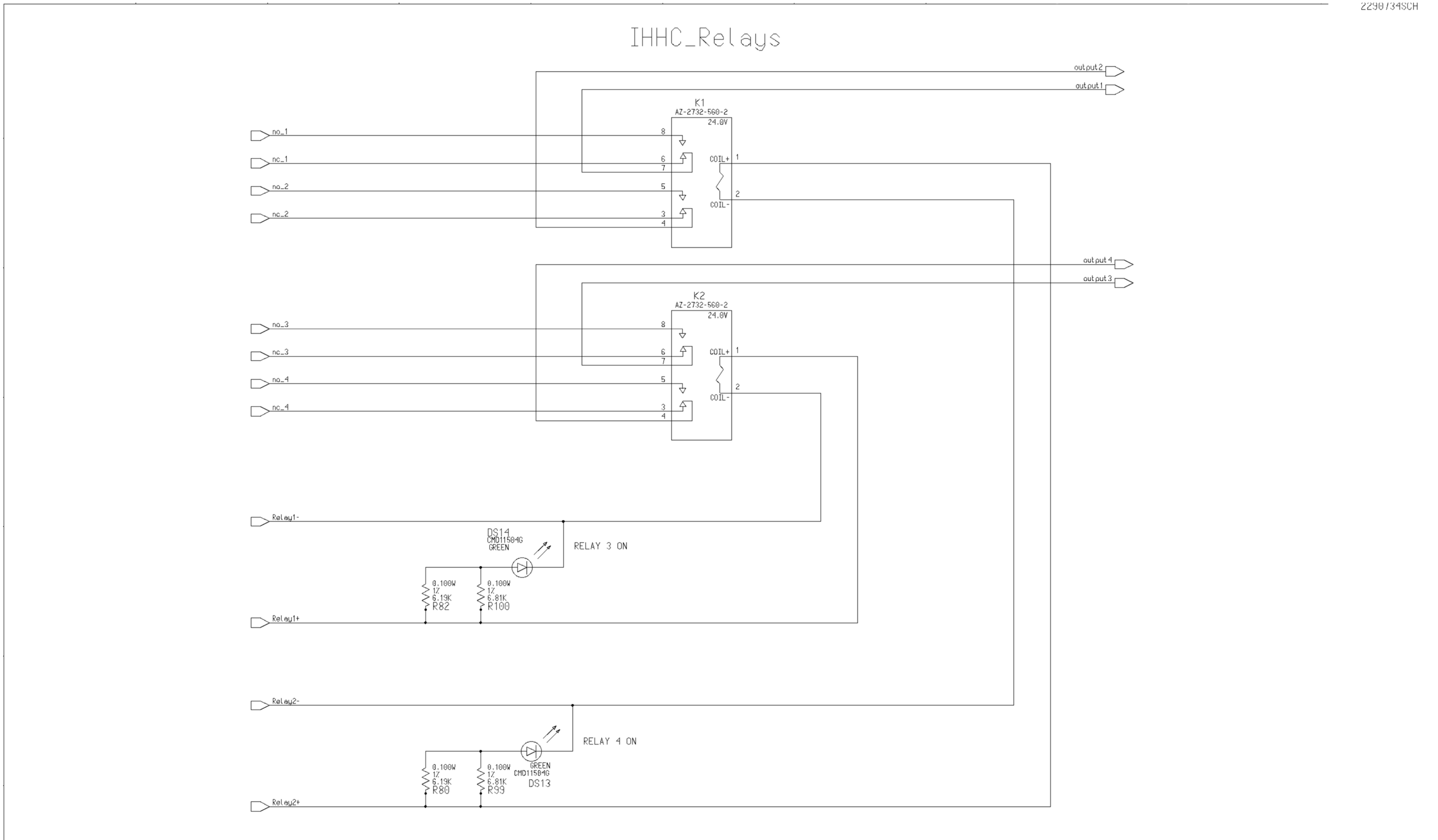


Figure 6-10 2290734, Sheet 9, Rev 0



BLOCK PATHNAME $GEMS_CT_PAT_POS/2290734_ASM/Sonata_Remote_Tilt/IHHC_relay		BLOCK SHT 1 OF 1		
REV PCN DESCRIPTION MADE BY DATE APPR BY DATE	MADE Michael Peete DATE 30/JAN/01 CHECK DATE	GE MEDICAL SYSTEMS MILWAUKEE, WI USA	DESIGN TITLE SONATA W/RTILT FIRST MADE FOR WARP 3	
REV PCN DESCRIPTION MADE BY DATE APPR BY DATE	APPR DATE OTHER DATE		DRAWING NO. 2290734SCH	REV 0 SIZE B SHT/CONT. ON 10 / 11

Figure 6-11 2290734, Sheet 10, Rev 0

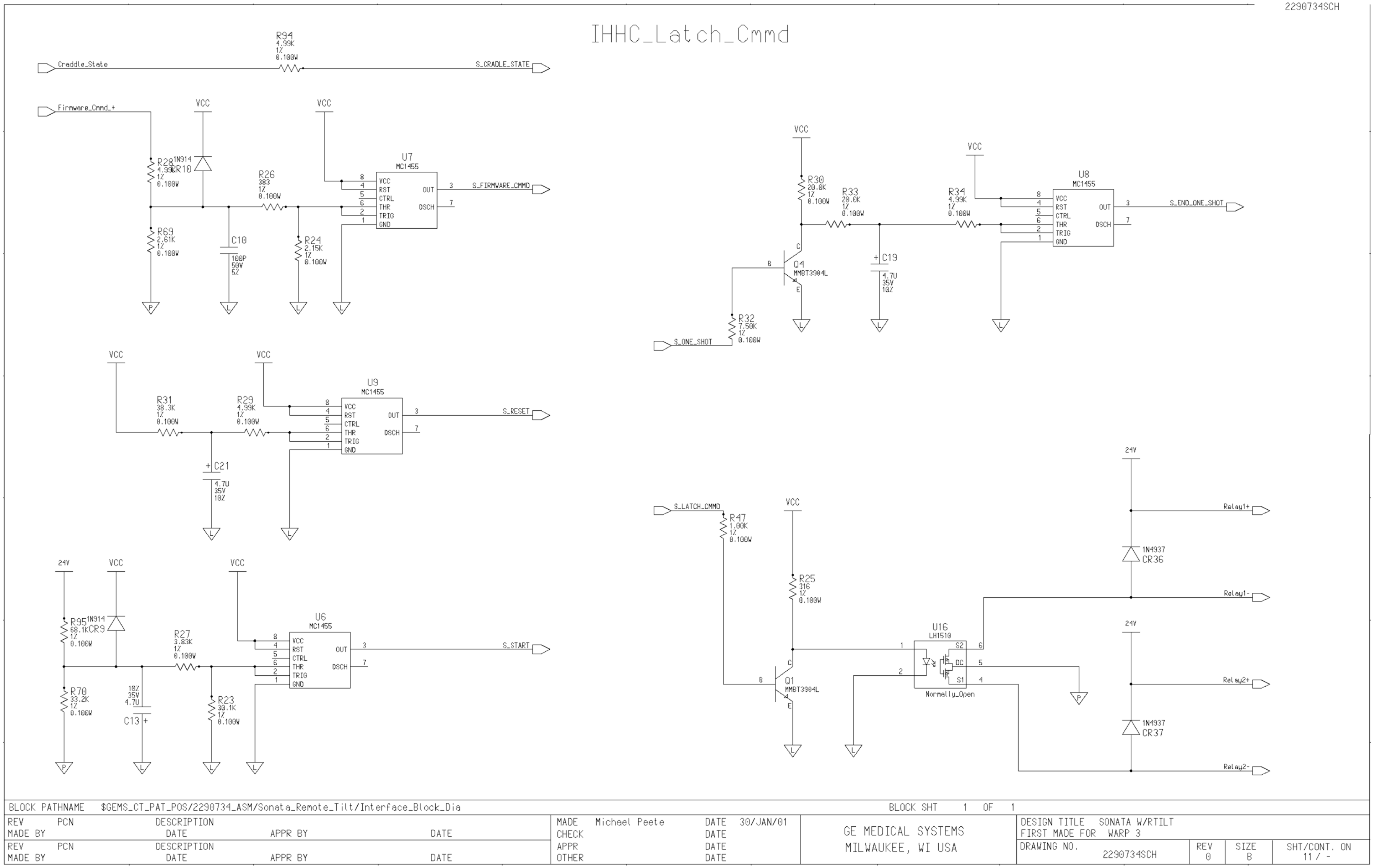


Figure 6-12 2290734, Sheet 11, Rev 0

Section 7.0
Replacement Parts

7.1 Remote Tilt For Installed Base (2300201)

MARK	NUMBER	QTY	FRU CODE	DESCRIPTION/NAME
1	2295327	1.000	N	REMOTE TILT UPGRADE KIT {*For further detail, see* Section 7.2, *beginning on* page 67.}
2	2299680-100	1.000	N	REMOTE TILT INSTALLATION MANUAL
3	2301300-100	1.000	N	REMOTE TILT OPERATOR MANUAL
4	2305935	1.000	N	KEY - REMOTE TILT SOFTWARE MOD
5	46-302200P12	3.000	N	RATING PLATE

Table 7-1 Remote Tilt for Installed Base Part List

7.2 Remote Tilt Upgrade Kit (2295327)

MARK	NUMBER	QTY	FRU CODE	DESCRIPTION/NAME
1	46-264368G3	1	1	ENHANCED TABLE CONTROL BOARD
2	2169590-2	1	N	CR INTERCOM/PRESCRIBED TILT BOARD {*For further detail, see* Section 7.3, *beginning on* page 68.}
4	2286908	1	1	TILT SENSOR ASSEMBLY - FRONT
5	2286909	1	1	TILT SENSOR ASSEMBLY- REAR
6	2286911	10	N	SCREW, TILT SENSOR MOUNTING
7	2295228	1	1	KIT CABLE ADAPTER #1
8	2295230	1	1	KIT CABLE ADAPTER #2
9	2295231	1	1	KIT CABLE ADAPTER #3
10	2295232	1	1	KIT CABLE ADAPTER #4
11	2295233	1	1	CABLE - TABLE
12	2295234	1	1	CABLE - FRONT COVER
13	2295235	1	1	CABLE - REAR COVER
14	2139577	1	1	CABLE - GANTRY TO CONSOLE
15	2290734	1	1	SANOTA BOARD WITH REMOTE TILT
16	2295236	1	N	SENSOR MOUNTING DRILL WITH GUIDE
21	46-208758P1	35	N	CABLE TIE, 3.62" X 0.094"
22	46-170127P5	1	N	MASKING TAPE, 1.0" WIDE, LIGHT TAN
23	46-170619P1	1	N	RTV SILICON RUBBER ADHESIVE
24	2271254	1	1	CABLE, RIBBON - ETC BOARD J5 TO SANOTA BOARD J5

Table 7-2 Remote Tilt Upgrade Kit Part List

MARK	NUMBER	QTY	FRU CODE	DESCRIPTION/NAME
25	2271255	1	1	CABLE, JUMPER - ETC BOARD J16 TO SANOTA BOARD J3
26	2299221	1	1	CABLE, RIBBON - ETC BOARD J4 TO SANOTA BOARD J4
27	46-208984P17	2	N	BRASS STANDOFF, MALE/FEMALE, 6-32
28	46-208560P68	2	N	SCREW, BINDING HEAD, 6-32 X 0.25"
29	46-328433P2	10	N	WASHER LOCK - EXTERNAL 4.3 MM
30	46-328425P1	10	N	NUT HEXAGON 4 MM"

Table 7-2 Remote Tilt Upgrade Kit Part List (Continued)

7.3 Intercom/Prescribed Tilt Board Assembly (2169590-2)

MARK	NUMBER	QTY	FRU CODE	DESCRIPTION/NAME
1	2167014	1.000	1	RHAPSODE INTERCOM BOARD
2	2268979	1.000	N	BRACKET - CIRCUIT BOARD
3	46-208560P48	6.000	1	SCREW
4	1000904P279	6.000	1	ID 0.1919 OD 0.3125 THK
5	46-287187P4	2.000	N	SCREW, SLT HD FLT, 2.5 x 6.0 MM
6	46-221417P5	1.000	1	FEMALE SCREWLOCKS
7	46-170686P3	AR	N	LOCTITE 242, 5OCC
8	2290734	1.000	1	SANOTA BOARD WITH REMOTE TILT
9	46-170015P41	6.000	1	SCREW

Table 7-3 Intercom/Prescribed Tilt Board Assembly Part List

7.4 Keyboard w/SCIM Upgrade Kit (2275758)

There are five Keyboard with SCIM Upgrade Kits available. Each upgrade kit is determined by the catalog number based by language. The five languages, and their corresponding catalog number and kit part numbers are as follows:

LANGUAGE	CATALOG NUMBER	ASSEMBLY NUMBER
English	B7800KE	2275758
French	B7800KF	2275758-2
German	B7800KG	2275758-3
Scandinavian	B7800KS	2275758-4
Italian	B7800KH	2275758-5

Table 7-4 Keyboard w/SCIM Upgrade Kit Catalog and Assembly Numbers

MARK	NUMBER	QTY	FRU CODE	DESCRIPTION/NAME
1	2275756	1.000	1	KEYBOARD, NORTH AMERICAN
2	2275757	1.000	N	FUNCTION KEY TEMPLATE - ENGLISH
	2275757-2	1.000	N	FUNCTION KEY TEMPLATE - FRENCH
	2275757-3	1.000	N	FUNCTION KEY TEMPLATE - GERMAN
	2275757-4	1.000	N	FUNCTION KEY TEMPLATE - SCAND
	2275757-5	1.000	N	FUNCTION KEY TEMPLATE - ITALIAN
3	2275752	1.000	1	SCIM COLLECTOR - ENGLISH
	2275752-2	1.000	1	SCIM COLLECTOR - FRENCH
	2275752-3	1.000	1	SCIM COLLECTOR - GERMAN
	2275752-4	1.000	1	SCIM COLLECTOR - SCANDI
	2275752-5	1.000	1	SCIM COLLECTOR - ITALIAN
4	2262158-9	1.000	1	CABLE, 50 PIN SCIM KEYBOARD
5	2275872-100	1.000	N	SCAN CONTROL/INTERCOM MODULE

Table 7-5 Keyboard w/SCIM Upgrade Kit Part List

GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 414.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.

GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006